The Institute for Petroleum Research and Geophysics

General Aspects and Initial Results of Magnetotelluric Surveys in Israel

Alexander Klang

March 1970

Report No. 1044

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ABSTRACT

The magnetotelluric method considered as a depth sounding tool, enables to find the major conductive and resistive sections in the sedimentary column and can help determine the depth of the basement and map its surface.

The Institute for Petroleum Research and Geophysics decided to adopt this method as an additional tool in geophysical exploration in Israel and in the first successful experiment, in March 1969, twenty-two measurements were taken along a traverse Afiq 1 - Sharsheret 1, 2 - Ziqlag 1 - Zohar - Lot.

Three resistivity complexes were found in the sedimentary section which could be correlated only generally with the geologic formation penetrated by the wells.

The deep high resistivity basement situated at depths greater than those penetrated by wells is considered to be of Pre-Cambrian age or even older.

The first magnetotelluric survey conducted in Israel has proved the possibilities of this method as a geophysical prospecting tool.

Some difficulties were encountered regarding the recording system and the interpretation.

It is proposed to develop this method further and to use it as a routine tool in goephysical prospecting.

It is also proposed that the existing equipment be supplemented with the necessary units to enable it to record four or five components of the magnetotelluric field and that computer programs be acquired to enable computation with two dimensional anisotropic earth models.

GENERAL ASPECTS AND INITIAL RESULTS OF MAGNETOTELLURIC SURVEYS IN ISRAEL

INTRODUCTION

The magnetotelluric method may be considered a depth sounding tool which gives results in the form of what is essentially a greatly smoothed induction log showing the major conductive and resistive section in the sedimentary column. It can help determine the depth of the basement and map its surface; it can be a gross correlation tool in geophysical exploration to great depth because it looks at large scale everage properties, ignoring much of the disorganized fine detail. Knowledge of the deeper features of sedimentary basins and of the gross variations of electrical conductivity within a sedimentary basin can be of considerable importance to the basic understanding of structure and stratigraphy, and therefore to the exploration for oil.

The Institute for Petroleum Research and Geophysics (IPRG) decided to adopt this method as an additional tool in geophysical exploration in Israel, and has been conducting experiments and perfecting techniques and instrumentation since 1965. The IPRG is in close contact with Centre de Recherches Geophysiques, France, where experimental instruments have been developed and with Geoscience Incorporated in the United States where this method is now being developed for commercial and industrial uses.

ERRATA

Page 8, formula : $\rho_a = 0,2\ T\left[\frac{Ex}{Hy}\ 2\right]$

should be : $\rho_a = 0,2\ T\left[\frac{Ex}{Hy}\right]^2$

Page 14, line 15 : 100 cm

should be : 100 Ω cm

Page 14. line 21 : Mskhtesh

should be : Makhtesh

THE MAGNETOTELLURIC METHOD

Basic Theory

The relationship between the natural alternating electric and magnetic fields measured at the surface of the earth has long been studied as a topic of intrinsic scientific interest and, more recently, as a means of determining the electrical conductivity distribution in the earth.

In 1868 G.B. Airy made the first coordinated study of earth currents and their relationship to magnetic variation. Tereda (1919) appears to have been the first to measure the dependence of the relationship between electric and magnetic components on the conductivity of the ground. He demonstrated that the ratio of an orthogonal electro-magnetic pair, (Ex to Hy) was inversely proportional to the square root of the period of fluctuation.

A first attempt for a systematic use of the earth telluric currents was made by Conrad Schlumberger in 1912. From 1950, when Tichonov suggested using natural electro-magnetic field for crustal sounding to 1953, when Cagniard published his classic paper, the idea of using these fields (called magnetotelluric by Cagniard) as a practical means of determining subsurface conductivities was developed.

Natural earth currents are strongly associated with fluctuations of the magnetic field above the surface of the earth. These fields originate over 100 Km. above the earth surface, presumably in the ionosphere or magneto-sphere, and propagate via a multitude of mechanisms to the earth surface.

Essentially the magnetotelluric method involves the comparison at the same time and place, of the horizontal components of the magnetic and electric fields associated with the flow of telluric currents. In fact, the electric field at the surface is dependant on the averaged conductivity from the surface to the depth of penetration for the particular frequency of the field being considered. This property allows us to obtain estimates of the electrical conductivity as a function of depth. This is the main parameter of the method.

It is important to state that quantitative interpretation is only possible, in a practical sense, for a model which assumes a horizontally layered earth and that the sources of the primary fields are sufficently far away so that the fields may be assumed to consist of uniform plane waves. In the magnetotelluric analysis proposed by Cagniard (1), a resistivity profile of a homogeneous stratified earth model is determined from the ratio of orthogonal tangetial components of the natural electric (Ex) and magnetic (Hy) fields, at the surface of the earth. Thus an apparent resistivity may be computed from: $\rho_a(T) = 0.2\ T \left[\frac{Ex}{Hy}\right]^2$, where T is the period of the particular frequency of the field being considered. It is important to note that the Cagniard model requires resistivity values which are invariant with the orientation of the measuring axis. From field observations this condition is sometimes met, but most often an orthogonal set of measurements yields considerably different resistivity estimates under rotation of the measuring axes.

These shortcomings of the Cagniard model were recognized by Cantwell (2) and others and a more realistic anisotropic model was proposed. This model involves the measure simultaneously of north-south (Ex and Hx) and east-west (Ey and Hy) components of natural fields over a range of frequencies.

Instrumentation

Although the theory of the magnetotelluric method is well established in the literature, the practical application of this method is limited due to difficulties encountered in the assembly of correct instrumentation (recording systems).

According to the magnetotelluric method, for the Cagniard model, the orthogonal tangential components of the natural electric (Ex) and magnetic (Hy) fields at the surface of the earth are simultaneously measured for a period T, and these measurements must be repeated for different periods T.

The depth of penetration attained by this method depends on the resistivity of the layers and on the period T which could be sensed by the magnetotelluric field recording.

The intensities of the magnetic and telluric fields vary with the period T and reach very low values for a definite range of periods (fig. 1, 2.). From these considerations the characteristics which must be achieved in the recording system can be envisaged. The equipment must be built for a very wide frequency range (0.002 - 150 Hz) to enable the recording of the magnetic and telluric events at all the periods considered necessary for the area of interest.

The equipment must be capable of a very narrow filtration range when the recorded magnetic and telluric events are played back and displayed for interpretation.

The equipment must permit a reliable reading of the magnetotelluric events (the amplitudes of the magnetic and telluric fields) for all the frequency ranges of interest. Due to the variation of the signal strength with the frequency, the equipment must allow a wide range of distortionless amplification of the signal.

Because of these specific characteristics the assembly of a good recording system is difficult; at present there is no standard equipment for magnetotelluric prospecting.

Due to the above considerations and because of lack of experience with this method in Israel and the lack of acquaintance with any type of magnetotelluric recording equipment, the Institute contacted the "Centre d'Etudes Geophysiques" in France, and the author of the present Report visited that Institute in November 1965 to study their equipment.

The Recording System

The Recording System used in Israel is the same as that used in France by the "Centre d'Etudes Geophysiques" (Fig. 3)

The system is built partly of French and partly of U.S. components.

The magnetic detector (magnetometer)(Fig. 4) is 2.20 m long, with a diameter of 10 cm. It consists of 600,000 - 9000,000 windings with a core of Mo-Metal (with Molybdenum). This detector has a good sensitivity and can detect small variations in the magnetic field in the frequency range employed in magnetotelluric surveying. It is provided with an amplifier to allow a first stage amplification of the low level signal.

The telluric detector is the usual telluric line, with two porous-pot electrodes which are planted in the ground about 600 m apart and connected by insulated leads to the input of the recording system.

The signal received by the magnetic and telluric detectors is passed through a three-stage amplifier with calibrated gain. Included in the amplifier are notch filters for 50 Hz and its third harmonic of 150 Hz. (fig 5).

In the amplifier a system of band pass filters (combination of high-cut and low-cut filters) is employed for pre-filtering the signals. The signals are thus not disturbed by noise during amplification.

The amplified signals, on passing through the interconnecting panel, may be treated as follows:

1. The signals are passed through the filters (KRONHITE 335 (R)) (Fig. 5) for narrow band filtration and recorded on paper on the 2 channel chart-recorder (MASSA-BSA 260).
2. The signals are recorded directly on paper on the chart-recorder (MASSA-BSA 260) and on magnetic tape on the tape-recorder (AMPEX-SP 300) (Fig. 6,7,). This is a 4-channel recorder; at present the Institute uses 2 channels only.
3. The signals are recorded directly on paper on the chart-recorder.
4. The signals are passed through the filters and recorded on tape by the tape-recorder. The chart-recorder simultaneously records the signal of known frequency to obtain the base of time.

By using the interconnecting panel, the magnetic tapes of the tape-recorder can be played back in the following ways:

1. Through the filters and recorded by the chart-recorder,
2. Directly on the chart-recorder,
3. Through the filters again on tape.

A low frequency signal generator is used for calibration, following each recording made at a definite period. Power is supplied by a generator.

Field Technique

The magnetometer and the telluric line are laid-out perpendicular to each other (Fig. 3A) . Field operations begin by recording the long periods of the magnetotelluric variations (Frequency band 1/5 Hz to 1/125Hz) on tape and for simultaneous checking on paper. Recording continues for several hours, during which time enough magnetotelluric events are obtained to permit a sufficiently accurate interpretation.

A calibration signal from the signal generator for a frequency chosen in the frequency band 1/5 - 1/125 Hz is then recorded on the tape. After this, recording continues for about an hour in the frequency range of 1/5 to 5 Hz. This operation is followed by calibration in the same way.

The operation is continued on the frequency band 2 - 120 Hz, by passing the signals through the filters and recording them directly on paper for each of the frequencies 4, 8, 16, 32, 45, 60, 80, 100 and 120 Hz. Each recording is followed by calibration in the same frequency range.

The recorded tapes are played back through the filters to the chart-recorder. In order to save time, the recorded magnetic tape is played back with a speed eight times greater than that used in the recording. Each operation is followed by calibration in the same frequency as that used in playback.

The frequencies used for playback of the frequency range 1/5 Hz to 1/125 Hz are:

1/5 Hz
1/10 Hz
1/20 Hz
1/30 Hz
1/60 Hz
1/120 Hz
1/200 Hz (outside the range)

The frequencies used for playback of the frequency range 1/5 to 5 Hz are:

1/10 Hz (outside the range)
1/5 Hz
1/3 Hz
1/2 Hz
2 Hz
4 Hz

The Ampex-Sp 300 tape-recorder used in our equipment proved unreliable in operation. It introduces some noise which can disturb the signal and in playback the speed is not high enough to permit a reasonable playback time for digitizing the data for computer processing.

METHOD OF INTERPRETATION

The Cagniard Model

In the method of interpretation proposed by Cagniard (1) a resistivity profile of a homogeneous stratified earth model is determined from the ratio of orthogonal tangential components of natural electric, (Ex) and magnetic (Hy) fields, at the surface of the earth.

Thus, an apparent resistivity may be computed from:

$$\rho_a = 0{,}2\ T \left[\frac{Ex}{Hy}^2\right]$$

where:

ρ_a = The apparent resistivity in Ωm

T = The period in sec.

Ex = The telluric field component in a direction in mV/Km.

Hy = The horizontal magnetic field component in perpendicular direction in γ

} for the same period T

The records obtained directly by playing back the magnetic tapes for each period T, with the signal of calibration at the end of record, are analysed individually and all the magnetic and telluric events which have a good correlation on magnetic and telluric channels are considered. The amplitude ratio of the electric and magnetic events are calculated and an arithmetic mean is obtained. Using the amplifier and calibration coefficients of the channels, the true mean value $\frac{Ex}{Hy}$ is obtained.

From the formula: $\rho_a = 0{,}2\ T \left[\frac{Ex}{Hy}\right]^2$ the value of the apparent resistivity ρ_a, for the considered period, T, is obtained.

This operation is repeated for all the periods recorded and the corresponding values of ρ_a are obtained.

By plotting these values on paper, on bilogarithmic coordinates $\rho_a - \sqrt{T}$, a curve for the field station recorded can be drawn and a model of resisitivity and thickness values can be computed or constructed by matching this curve with standard master curves.

By using a computer program for a multilayer case, the theoretical curve for the considered model can be constructed and by fitting this theoretical curve with the experimental curve the proposed interpretation of the resisitivity model is verified. If the theoretical curve for the compiled model does not fit the measured curve, the interpretation must be verified with a new model of resistivities and thickness values. This new model is again checked with the computer program.

Interpretation is completed when the theoretical curve coincides with the measured curve.

The theoretical studies of Prof. L. Cagniard have demostrated that there is a phase difference between the magnetic and telluric fields, the first being retarded in relation to the second. Cagniard has computed a set of phase difference curves versus period for various resisitivity contrasts for a two layer earth model. The use of this criterion in interpretation would help produce more accurate results. However, the difficulties encountered in assembling a reliable system for the recording of phase differences have so far been insurmountable ; at present, phase difference cannot be applied in the Cagniard model.

The Impedance Tensor Model

The Cagniard model requires resistivity values which are invariant with the orientation of the measuring axis It assumes a homogeneous horizontally stratified earth model which is obviously inadequate for many places.

From field observations it is known that most often an orthogonal set of measurements yields considerably different resistivity estimates under rotation of the measuring axes.

The impedance tensor model tries to avoid these shortcomings; it is based upon a rank two impedance tensor which takes into account the possibility of the effects of two dimensional lateral anisotropies and inhomogeneities.

For this model the north-south (Ex and Hx) and east-west (Ey and Hy) components of the natural electric and magnetic fields are measured over a range of frequencies (from about 0.0001 to 20 Hz).

The basic tensor model satisfies the two relations:

$$Ex = Zxx.Hx + Zxy.Hy$$

$$Ey = Zyx.Hx + Zyy.Hy$$

where Ex, Ey, Hx and Hy are the components of the field and Zxx, Zxy, Zyx and Zyy are the impedance elements.

Swift (9) has expressed the impedance elements in terms of auto powers and coherencies of the measured elements which can be written as:

$$Zxx = \frac{|Ex|}{|Hx|}\left(\frac{Coh\ ExHx - Coh\ ExHy\ Coh\ Hy\ Hx}{1 - |Coh\ Hx\ Hy|^2}\right)$$

$$Zxy = \frac{|Ex|}{|Hy|}\left(\frac{Coh\ ExHy - Coh\ ExHx\ Coh\ Hx\ Hy}{1 - |Coh\ Hx\ Hy|^2}\right)$$

$$Zyx = \frac{|Ey|}{|Hx|}\left(\frac{Coh\ EyHx - Coh\ Ey\ Hy\ Coh\ Hy\ Hx}{1 - |Coh\ Hx\ Hy|^2}\right)$$

$$Zyy = \frac{|Ey|}{|Hy|}\left(\frac{Coh\ EyHy - Coh\ EyHx\ Coh\ Hx\ Hy}{1 - |Coh\ HxHy|^2}\right)$$

where $|Ex| = (\langle Ex\ Ex^* \rangle)^{\frac{1}{2}}$

$$Coh\ Ex\ Hy = \frac{\langle Ex\ Hy^* \rangle}{(\langle Ex\ Ex^* \rangle \langle Hy\ Hy^* \rangle)^{\frac{1}{2}}}$$

and * represents the complex conjugate.

Where resistivity does not vary laterally apparent resistivity will be independent of location as well as of the actual directions of the measured components.

In that case Zxy = -Zyx = Z and Zxx = Zyy = 0 and the Cagniard model is applicable.

Where resistivity does vary laterally (lateral anisotropy) then in general Zxy ≠ Zyx and neither Zxx nor Zyy is zero. Then four different apparent resistivities $\rho ij = 0.2.T.Zij$ $(i = x, y)$ $(j = x, y)$ can be calculated, of which ρ xy and ρ yx are closest to scalar resistivities in their significance.

To find the proper directions of the anisotropy axes the rotation criteria which involve either the maximization of Zxy - Zyx or what is equivalent to minimazation of Zxx + Zyy are used. The values of ρ ij and Zij are then recomputed as if they had been measured in the proper direction to begin with.

The direction in the horizontal plane of the larger apparent resistivity (azimuth) is either parallel or perpendicular to strike, if strike is defined. Where strike direction changes with depth or with distance, the azimuth at a site will vary with frequency.

By using computer programs, all the calculated apparent resistivities mentioned before, and even plotted curves are obtained.

The interpretation involves the comparison of the field results with those obtained from computed models.

Two-dimensional (with lateral changes in one direction) model programs are known by now. By employing these programs (in one of which the vertical component H of the magnetic field is recorded) models used for the interpretation of the field results are computed. Because only two components (Ex and Hy) of the field could be measured with our recording system, we interpreted our data with the Cagniard model which as shown above, are not adequate for the real anisotropic and inhomogeneous earth model.

EXPERIMENTS

In the first successful experiment, in March 1969, twenty-two measurements were taken along a west-northwest, south - southeast traverse beginning at Afiq 1, continuing through Sharsheret 1, Sharsheret 2, Gerar 1, Ziqlag 1, the Zohar gas field, and terminating at Lot 1 well . Two of these measurements were discarded as unreliable (fig. 8)

RESULTS

The interpreted resistivity traverse (fig 9) shows three resistivity complexes in the sedimentary section; the top near surface complex about 200-400 m. thick, with resistivities of 220-800 Ω cm. (in some places 1500-1700 Ωcm)

The second complex about 4-6 Km. thick with resistivities of 10,000 - 30,000Ωcm. (sometimes 60,000 - 120,000Ωcm.) The lowest complex about 500-2000 m. thick with resistivities of 60-750Ωcm.

The lowest complex is followed by a very high resistivity formation which, in correlation with the final slope of the apparent resistivity curves is considered as representing the basement.

These three resistivity complexes are not continuous throughout the section, some of the discontinuities are marked by us. Some measurements at sites 9, 10, and 13 did not reach the basement and the last resisitivity value was from 8000 to 2250 Ω cm.

These results can therefore be correlated only generally with the geologic formations penetrated by the wells.

The first near surface resistivity complex represents the Paleogene-Neogene shale and clay formations with low resistivities.

The second complex represents all of the Cretaceous-Jurassic and may be the Triassic section. The high resistivity values may be the effect of the predominant dolomites and limestones; the interbedded thin marl and shale formations with their much lower resistivities are not distinguishable. The third resistivity complex situated at a depth of 3 to 7 Km. is deeper than the formations penetrated by the wells in the area. The deepest formations penetrated by the Zohar 8, Heimar 1, and Lot 1 wells was the Paleozoic. This low resistivity complex is considered representative of marl or high pressure shale formations in the Paleozoic. Very low resistivity values of less than 100 cm. are known for such shales in the literature (3).

The high resistivity basement cannot as yet be precisely discribed as to composition and age at this experimental stage. It is believed to be of Pre-Cambrian age or even older. Some wells have penetrated the Pre-Cambrian and found arkose (Ramon 1, Mskhtesh Qatan 2) or igneous rock (Kurnub 1). More magnetotelluric measurements are needed to examine the effect of the arkose and to see if the high resistivity igneous rocks which may form the basement (basalt, granite) can be differentiated from the arkose cover. A comparison of the interpreted resistivity traverse with the aeromagnetic maps shows some good correlation (fig. 10).

An interpreted trough axis is exhibited on the aeromagnetic map southeast of Sharsheret 2 well. This axis, if continued, would reach the area of site 12; this is evident on our resistivity section. At the same time, the lines showing a downward flexure on the aeromagnetic map are traced near magnetotelluric sites 22 and 14 and our section exhibits this.

The depth determined from magnetotelluric measurements cannot be verified with absolute accuracy with the depths calculated, from body shape anomaly for aeromagnetic survey, due to the quantitative limitations of the two methods.

In general it is considered that for a single magnetotelluric sounding the basement depth is determined with an accuracy of $\pm$ 10% in relation to the aeromagnetic depths. For example the depth to basement was placed at the Lot area at about 3600 m. and by the aeromagnetic map at 3300 m.

In the Sharsheret area at the trough axis the depth shown in the aeromagnetic survey is 5700 m. and it is about 8500 m. at point 12 in our survey, which may be the maximum depth of the trough.

The apparent resistivity values versus frequency were plotted for each magnetotelluric site and contoured line values are exhibited along the traverse (fig. 11.). No correlation could be obtained for magnetotelluric sites 7 and 8, and the apparent resistivity contour lines were interrupted in this part of the traverse.

A similar situation is observed at magnetotelluric sites 17 and 20, the apparent resistivity values being very different for the two sites. At the same time the apparent resistivity values for site 14 are much higher in the first part of the section breaking the continuity of the resistivity lines between sites 13 and 14.

CONCLUSIONS

The first magnetotelluric survey conducted in Israel has proved the possibilities of this method as a geophysical prospecting tool.

The magnetotelluric field can be measured in good conditions in Israel and interpretable results are obtained.

Data about the structure and depth of the basement were obtained and correlated well with the data obtained from the aeromagnetic survey.

Some difficulties were encountered regarding the recording system and the interpretation.

The recording system which was mounted at the Institute with components made in France and in the United States, with the advice of the Centre d'Etudes Geophysiques,France, did not prove to be reliable enough under field conditions. Performance was poor mainly at very low frequencies and stability was poor with calibrations not holding for more than short periods. Speed ranges are limited, the highest speed being 15 inch/second while a speed of at least 60 inch/sec would be required in order to make play-back times practical. The equipment allowed the recording of only two components (one of the electric field Ex and one of the magnetic field Hy) while complete magnetotelluric measurement and interpretation require the recording of four or even five components at the same time (two of the electric field Ex, Ey, and two or three of the magnetic field Hx, Hy, and even Hz). As a result, only the simple isotropic model of Cagniard could be used for the interpretation .

Computer programs could not be used for calculating the resistivities because of the analog form of recording and the incompatibility of the tape-recorder to be used for input to the A to D converter of the 360/44 computer.

RECOMMENDATION

It is proposed to develop this method further and to use it as a routine tool in geophysical prospecting. Considering the difficulties encountered with the first measurements and taking into account new development of this method in recent years we propose that the equipment be supplemented to enable the recording of four or five components of the magnetotelluric field; that the tape-recorder be replaced with a digital one reliable enough for local field conditions and compatible with the IBM 360/44 computer for data processing, and that computer programs be acquired for data processing with the use of two dimensional anisotropic earth model.

REFERENCES

1. L. Cagniard — Basic Theory of the magnetotelluric method of geophysical prospecting.Geophysics 18, 3, 605-635.

2. Cantwell T., — Detection and analysis of low frequency magnetotelluric signals. PhD. Thesis, Department of Geology and Geophysics, M.I.T. 1960

3. Walter H. Fertl. & Donald J. Timko — Occurence and significance of abnormal pressure formations , The Oil and Gas Journal 68, 1 97-108.

4. A. Klang — Magnetotelluric developments in France 1965, The Institute for Petroleum Research and Geophysics Report 1010/1966.

5. T. Meidav & A. Klang — General aspects and initial results of tellurometric experiments in Israel. The Institute for Petroleum research and Geophysics TM/390/65.

6. H.F. Morrison — The magnetotelluric method.Paper presented at Symposium: Application of natural electromagnetic fields in petroleum and mining exploration. March 19 - 23, 1968 University of California, Berkeley.

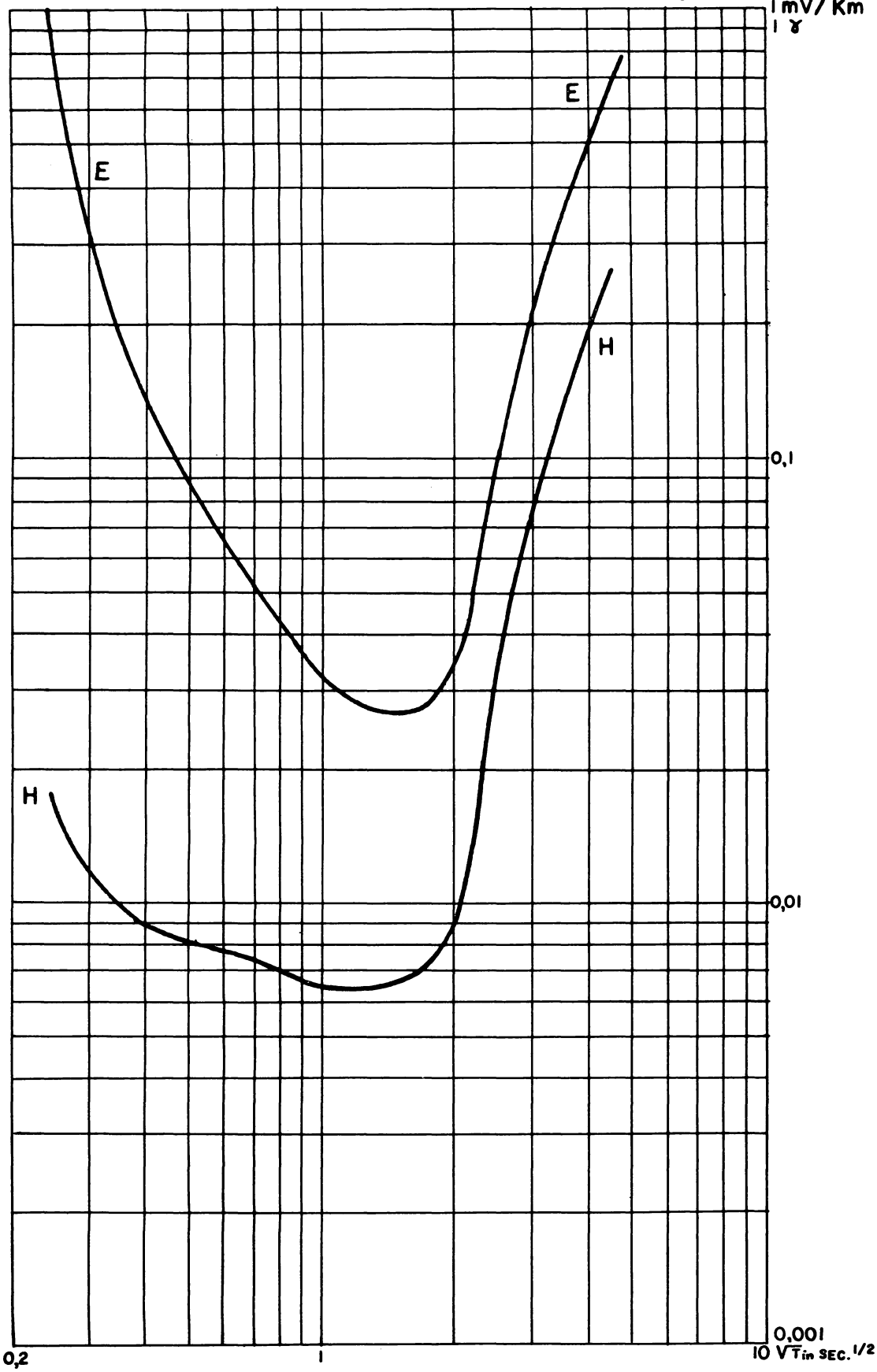
7. D. Rankin & I.K. Reddy — A magnetotelluric study of resistivity anisotropy.Geophysics 34, 3, 438 - 449.

8. H.W. Smith — Data processing for magnetotelluric. Paper presented at symposium; Application of natural electromagnetic fields in petroleum and mining exploration. March 19-23, 1968 University of California, Berkeley.

9. C.M. Swift Jr. — The nagnetotelluric method: low frequency field examples, Arizona and New Mexico. Paper presented at Symposium. Application of natural electromagnetic fields in petroleum and mining exploration. March 19-23, 1968 . University of California Berkeley.

10. K.H. Vosoff, H.Hasegawa &R.M. Ellis — Results and limitation of magnetotelluric surveys in simple geologic situation Geophysics 28, 5 778-792.

Fig. 1



Curves of E mv/km and H-γ as a function of $\sqrt{T}$ for a sedimentary column

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	PERIOD sec. / COLUMN	0,06	0,120	0,125	0,510	0,700	1,25	3	3,5	4,5	8	20
E mV/km	SEDIMENTS	0,980		0,184	0,051		0,030		0,030		0,161	0,650
	IGNEOUS ROCKS	2,80	2,77			0,228		0,685		0,945		6,55

Intensity of telluric events

	PERIOD sec / COLUMN	0,06	0,120	0,125	0,510	0,700	1,25	3	3,5	4,5	8	20
H γ	SEDIMENTS	0,018		0,0093	0,0076		0,0063		0,0076		0,050	0,266
	IGNEOUS ROCKS	0,0048	0,0054			0,0070		0,0265		0,020		0,305

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Intensity of magnetic events

Fig. 2

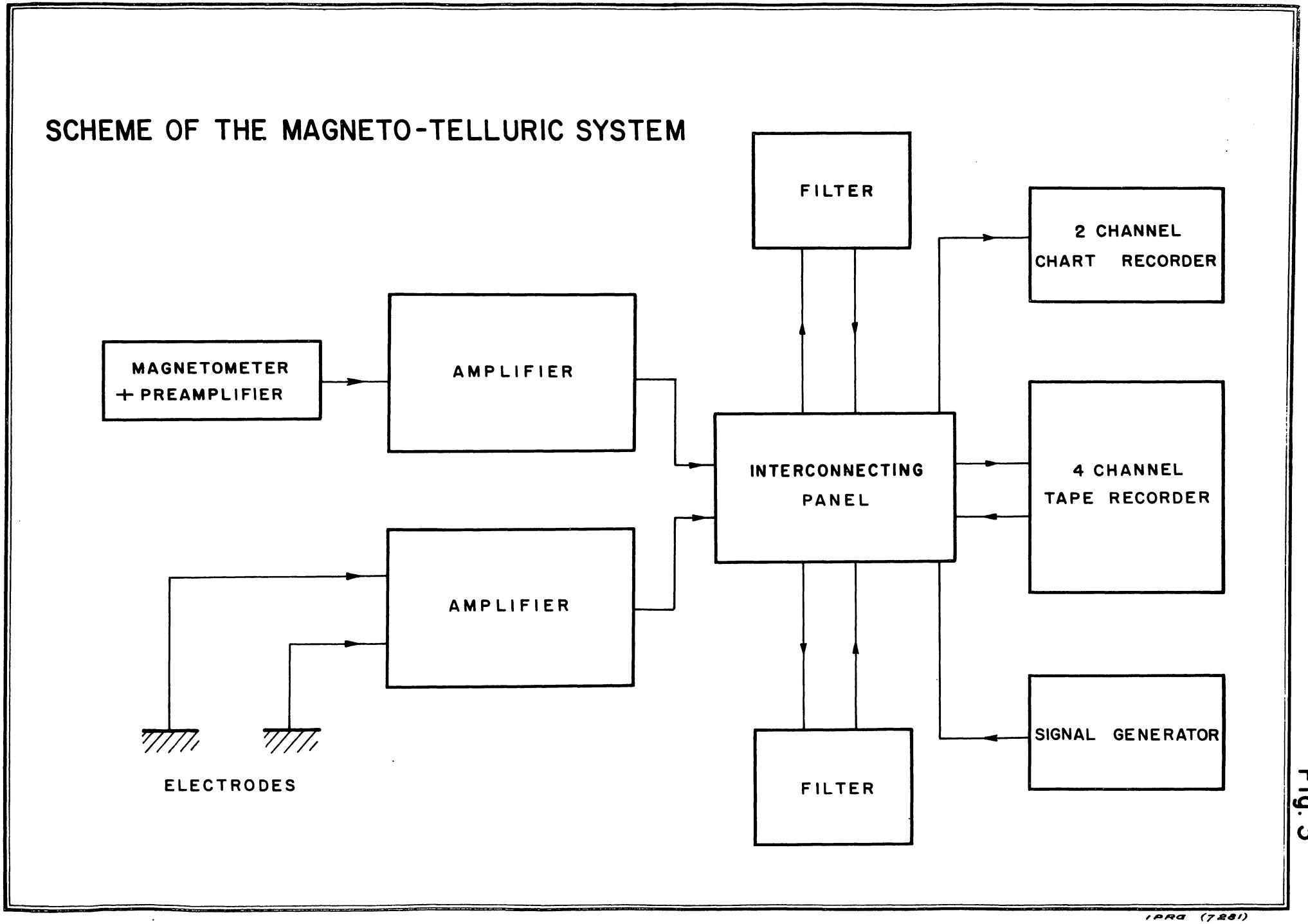
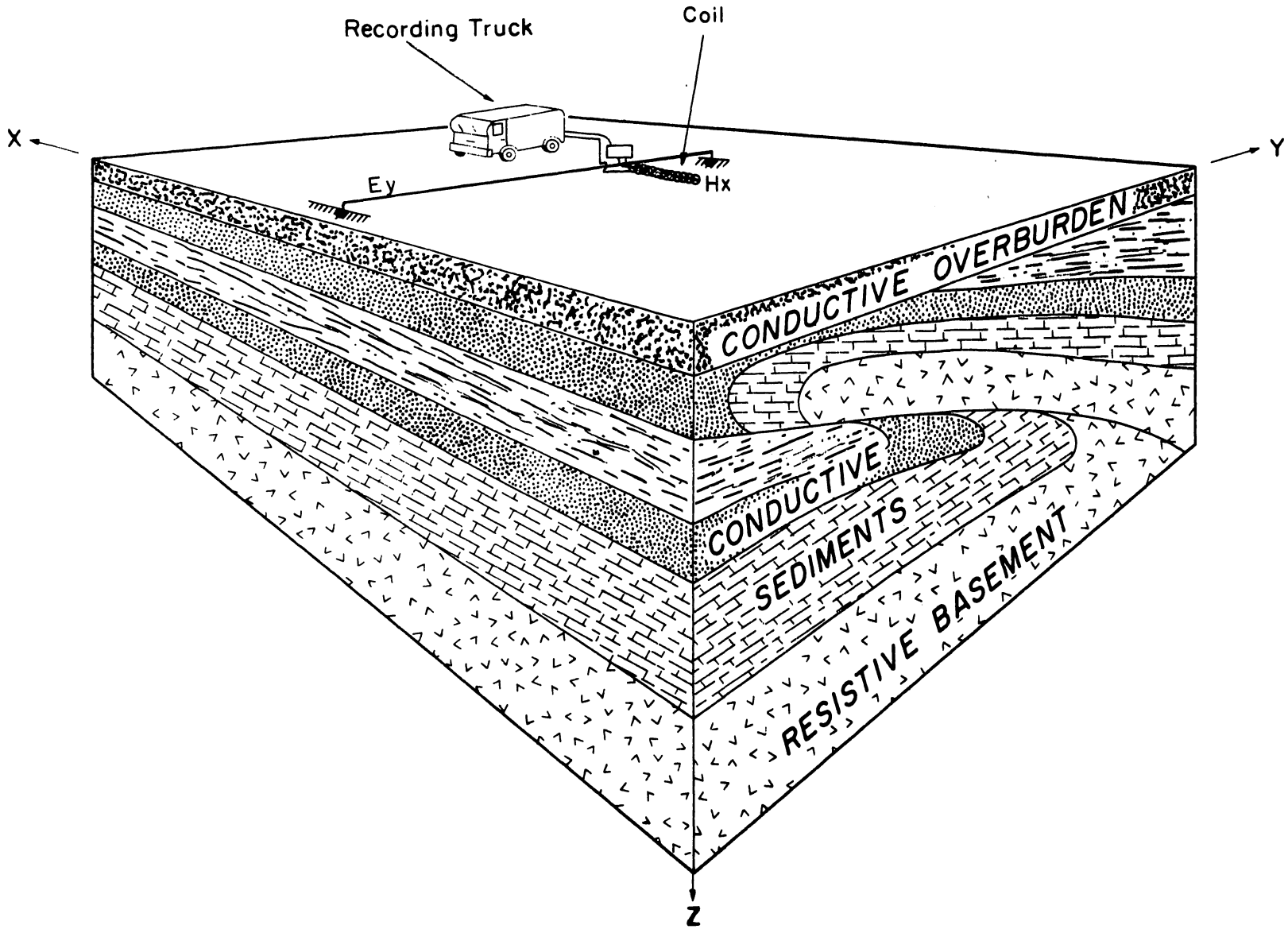


Fig. 3

IPRG (7281)



MAGNETO-TELLURIC FIELD SET-UP

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Fig. 3A

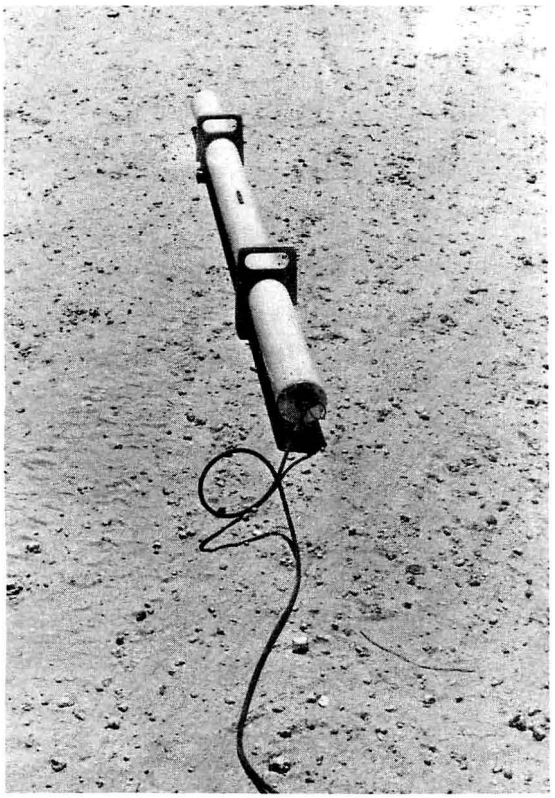


Fig. 4. The magnetic detector (magnetometer)

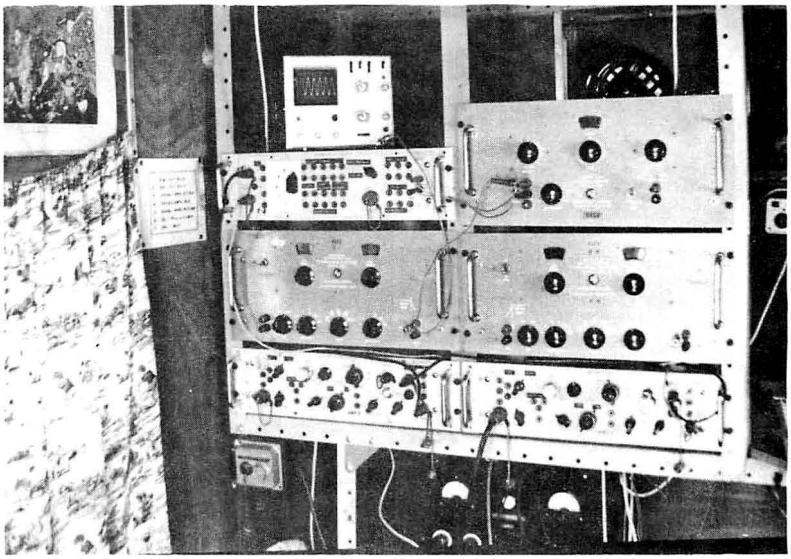


Fig.5. Front panel with the amplifiers filters signal generator

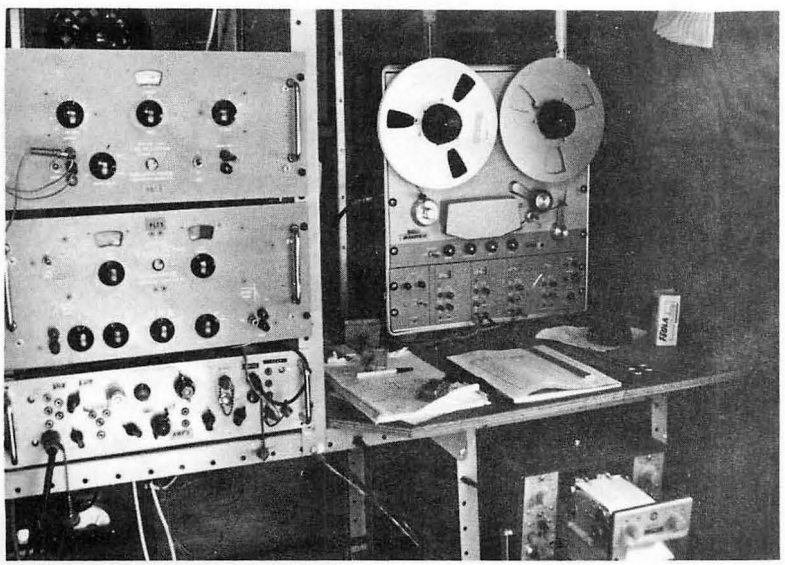


Fig. 6. Tape-recorder and Chart-recorder.



Fig. 7. Tape-recorder and Chart-recorder

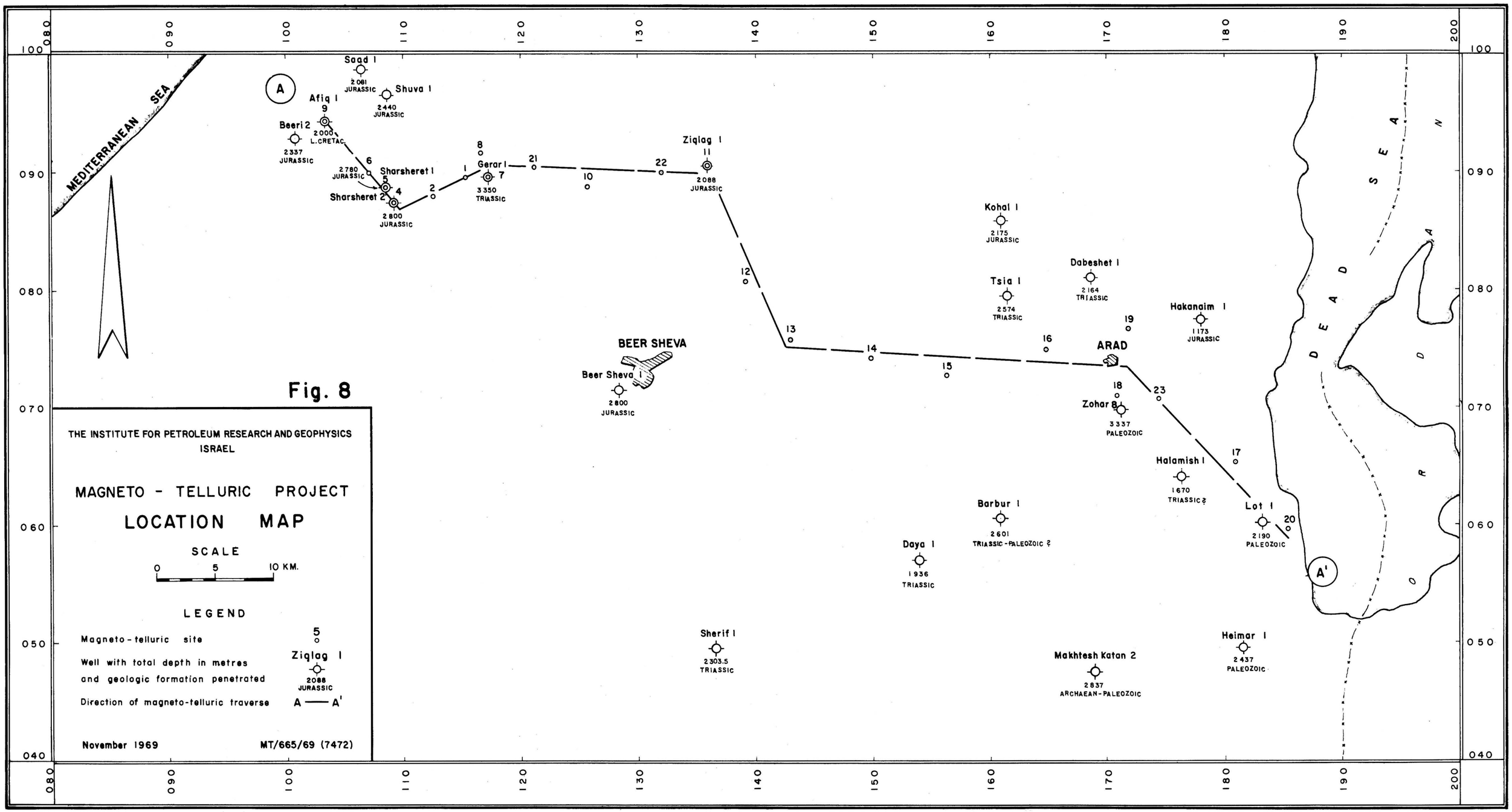


Fig. 8

THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS
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MAGNETO - TELLURIC PROJECT

LOCATION MAP

SCALE

0 5 10 KM.

LEGEND

Magneto-telluric site — 5

Well with total depth in metres and geologic formation penetrated — Ziqlag 1, 2088, JURASSIC

Direction of magneto-telluric traverse — A — A'

November 1969

MT/665/69 (7472)

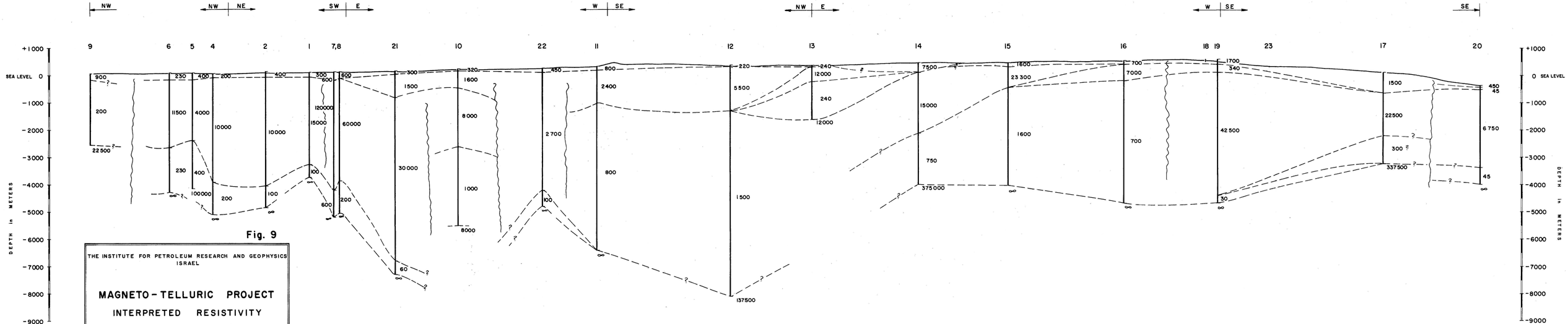


Fig. 9

THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS
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MAGNETO-TELLURIC PROJECT

INTERPRETED RESISTIVITY

TRAVERSE A–A'

SCALE

km. 2 1 0 1 2 3 4 5 km.

RESISTIVITIES IN OHM-CENTIMETERS

NOVEMBER 1969 MT/665/69(7473)

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MAGNETO - TELLURIC PROJECT

LOCATION MAP

AEROMAGNETIC MAP SUPERIMPOSED

SCALE

km. 0 1 2 3 4 5 km.

LEGEND:

- Magneto Telluric Site . . . . . 5 / 0
- Direction of Magneto-Telluric Traverse . . . . . A——A'
- Interpreted trough axis . . . . . o+o
- Basement -form lines- -bar- shows direction of down flexure . . . . . —x—

Estimated depth in meters below surface/sea level to the top of the -body-. Minus sign denotes elevation above sea level. 5700/5600

November 1969

Fig. 10

MT/665/69 (7474)

Joins Sheet 10-11

Joins Sheet 21 - 22

DEAD SEA

JORDAN

Yatta (Yuta)

Dhâhiriya

Cape Costigan

Cape Molineux

SALAD I.

AFIK I.

SHARSHERET 1

SHARSHERET 2

ZIQLAG 1

SHIKMIM 1

BEER SHEVA 1

CFAQIM

Deepening

5700/5600

4800?

4800/4400

4200?

3900/3100

3300/3100

TSIA 1

DABERET 1.

MESADA 1.

EINGEDI 2

KIDOD 1

ZOHAR

HALAMISH 1

BARBUR 1.

DAYA 1.

DIMONA

FAI 1.

HEMAR 1.

SDOM 1

SDOM 2.

Upper Mottled Zone (magnetite concretions)

Yeruham

Shuva

Zimrat

Zeruia

Mabbui m.

Eshbol

Gefahim

Sede Zev

Bet Yama

Bet Ha Gadd

Netivot

Sharsheret

Shaqeeda

Bahu

Beeri

Pa'ame Gashaz

Gashur

Patish

Ha Nas

Shemarya

Gerar

Shivtal

Urim

Ashan

Beersheva

Hazerim

Yatir

Ziqlag

Ramat Hanegev

Ezuz

Ammiaz

Hemar

Lot

Zohar

Mishmar

Zeelim

Ben Yair

Masada

Gannolim

Kidod

Arad

Amram

Hever

Ein Gedi

Dewid

Anuggot

Hazeva

Holed

Eshtemoa

Hevron

Adasha

Ter

Maon

Nabi

Samuel

Ashalim

Ere

Peratim

Amar

Hasa

Beit Sheva

Malhata

Yerucham

Retem

Aqeva

Ashan

Bror

Hagrar

Bussa

Lavan

Ashalim

Ammiaz

Peratim

Amar

Hasa

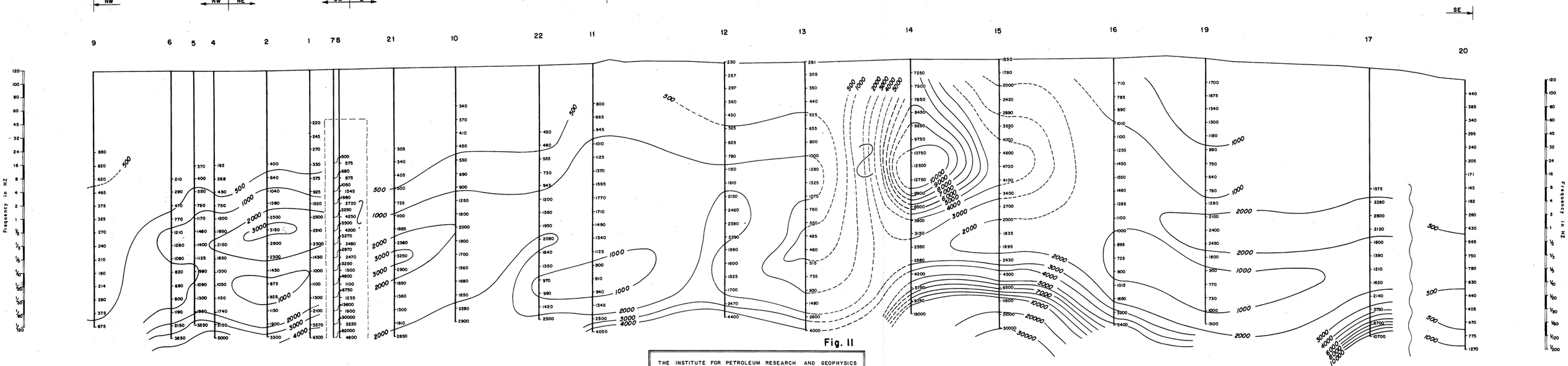


Fig. II

THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS
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MAGNETO - TELLURIC PROJECT

TRAVERSE A – A'

CONTOURED APPARENT – RESISTIVITY LINES

SCALE

km. 2 1 0 1 2 3 4 5km.

RESISTIVITIES IN OHM – CENTIMETERS

NOVEMBER 1969

MT/665/69 (7474)

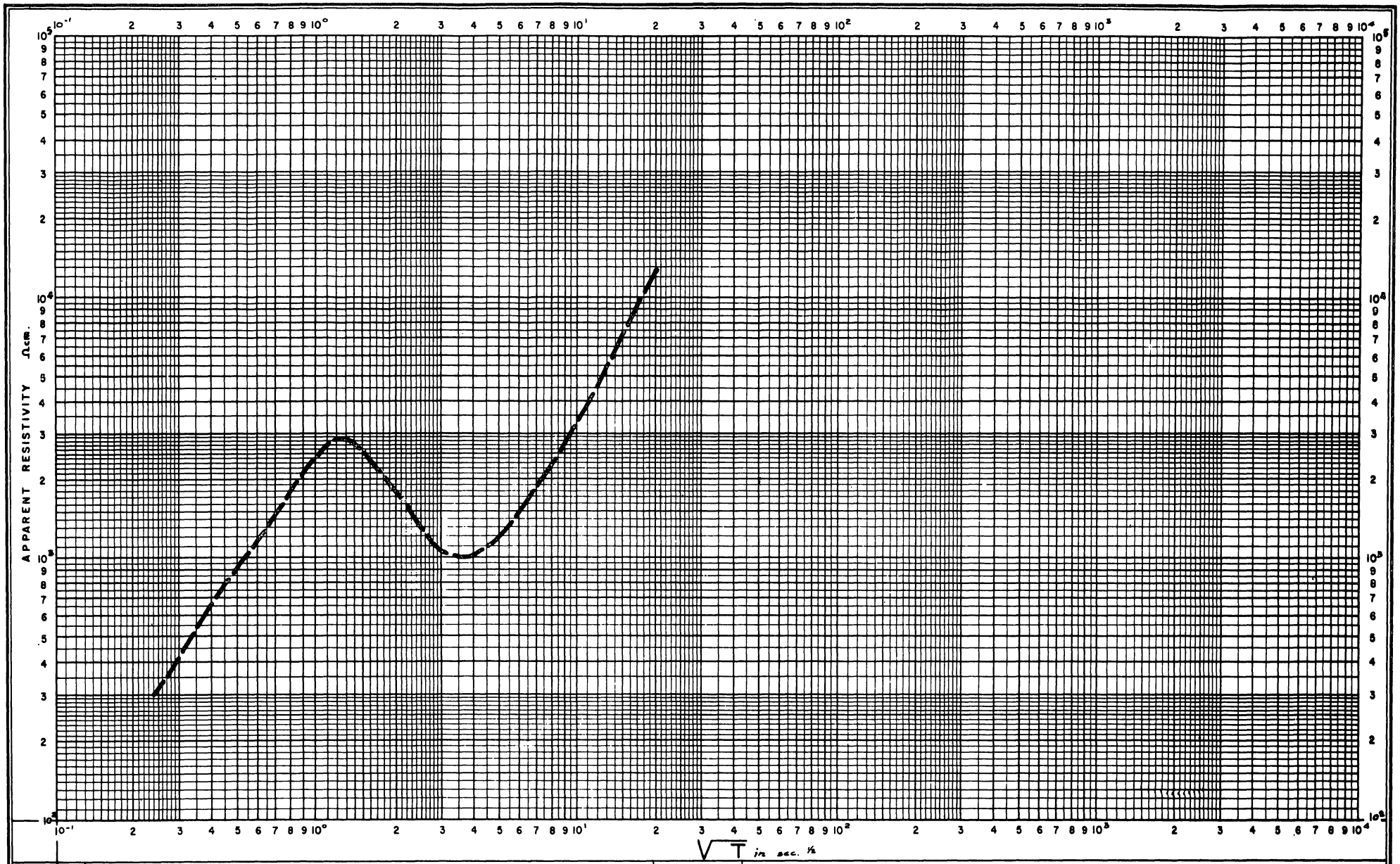
CONTOUR INTERVAL

1. Up to 1000 Ω cm. every 500 Ω cm.
2. From 1000 to 10000 Ω cm. every 1000 Ω cm.
3. From 10000 Ω cm. every 10000 Ω cm.

Appendix

Magnetotelluric Apparent Resistivity Curves

1, 2, 4—17, 19—22



M.T.P. No. 1. . . .

FILE No. . . . . .

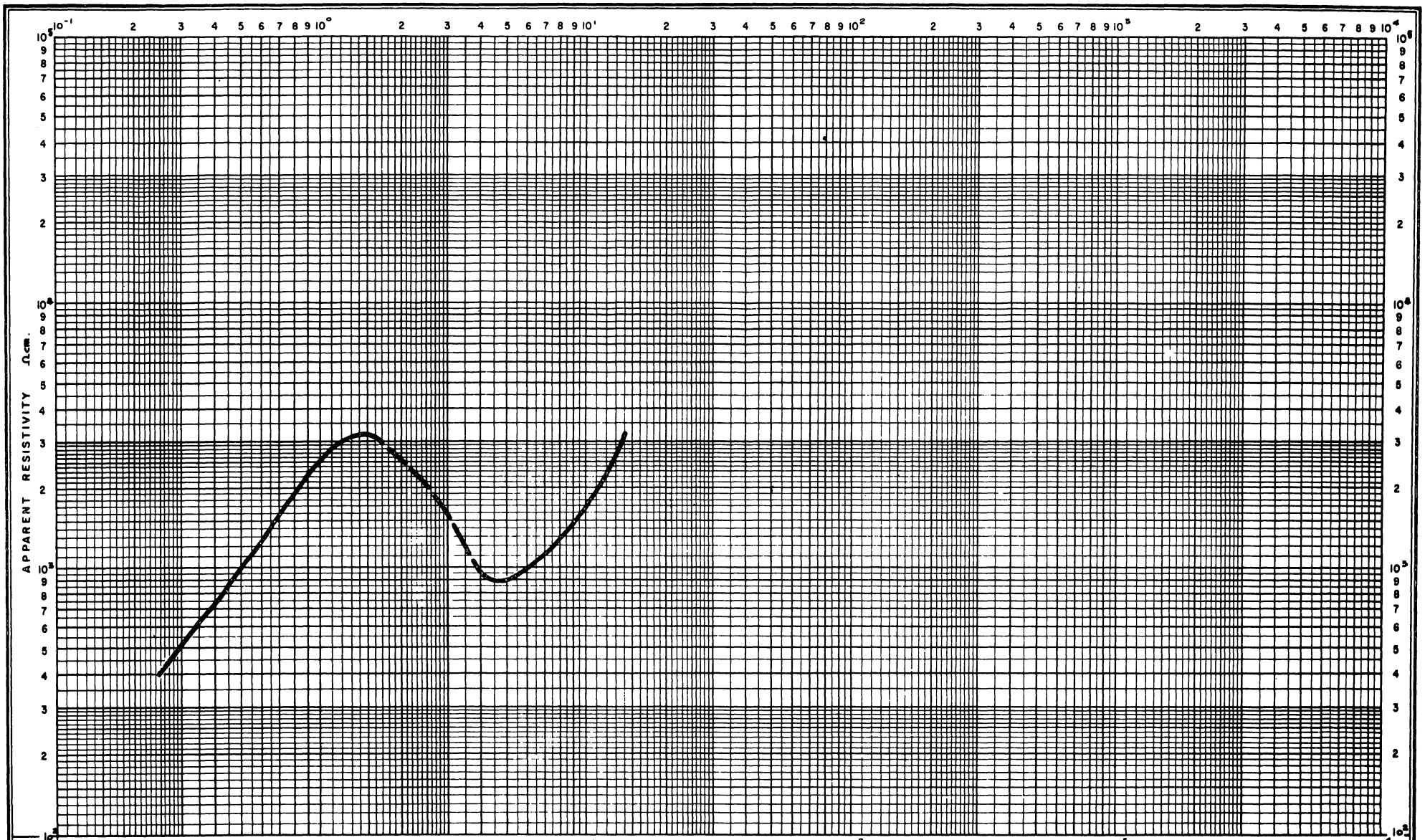
JOB No. MT/665/69. (7665)

LOCATION: GILAT

COORDINATES: 115250 — 089600

ELEVATION: 140 m.

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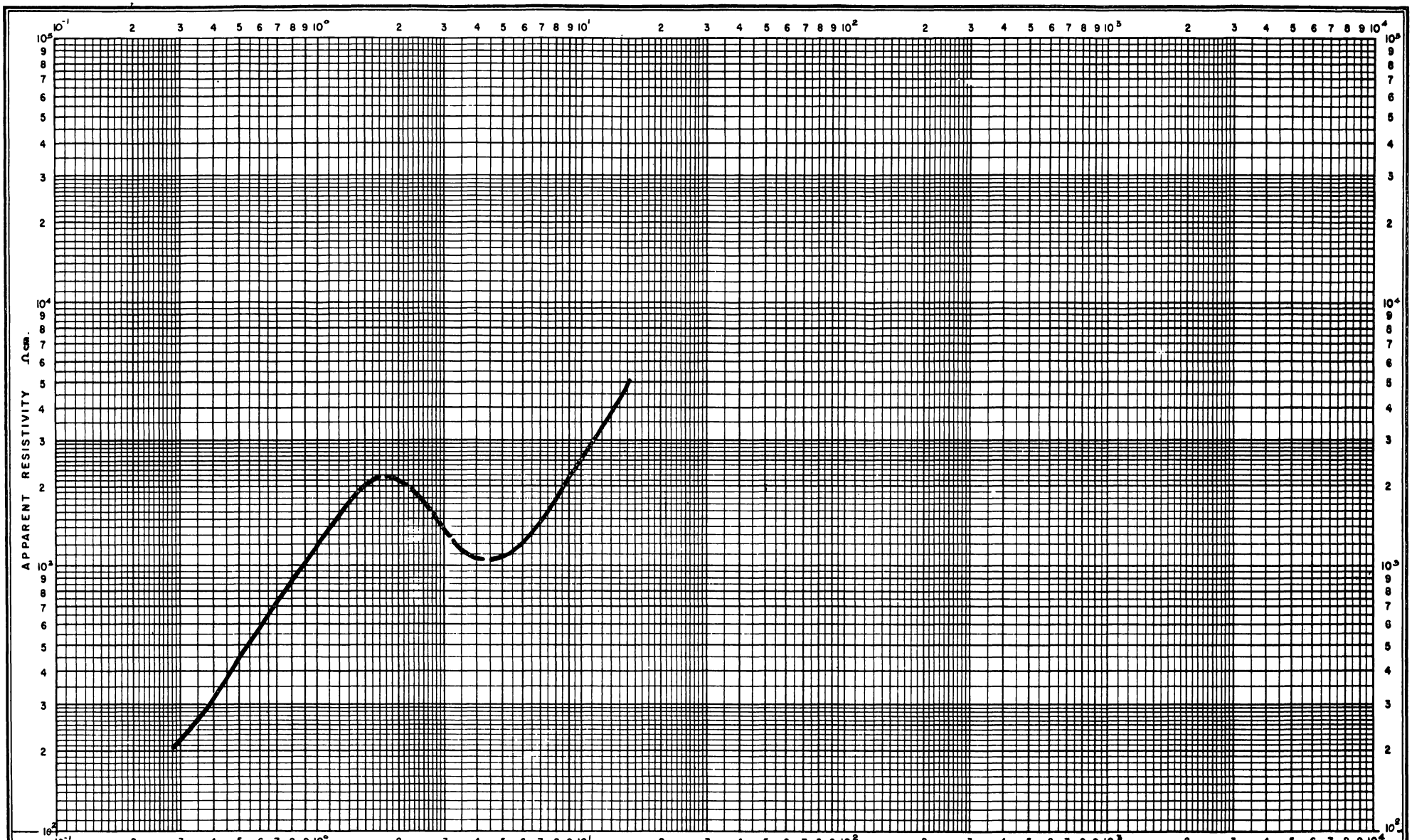


CALC. RES. Ωcm.	400	10000	100	∞
Depth in m.	200	4200	5000	

M.T.P. No. 2 . . .
FILE No. . . . . .
JOB No. MT/665/69.(7686)

LOCATION: GILAT
COORDINATES: 112500 — 087950
ELEVATION: 128 m.

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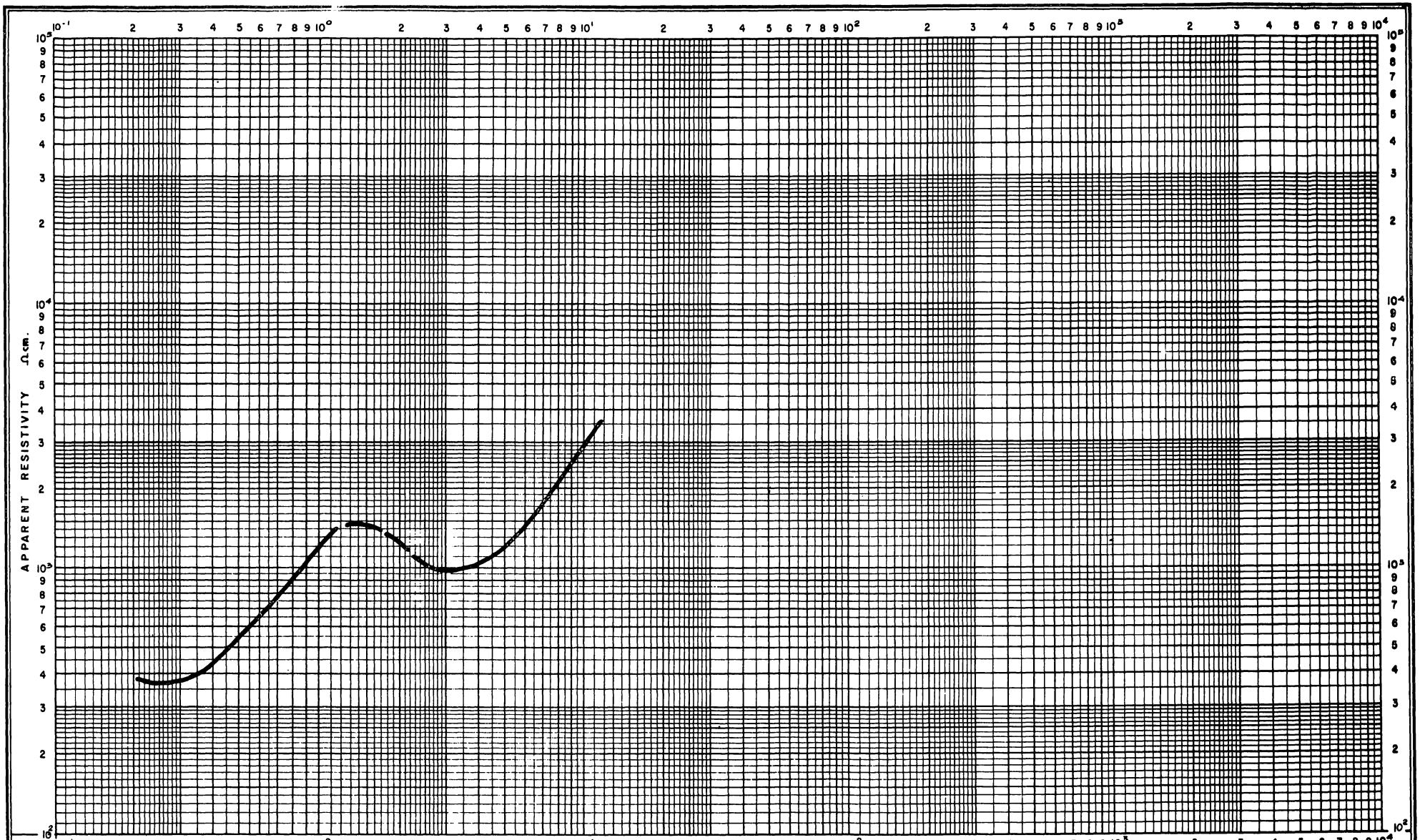


CALC. RES. Ωcm.	200	10000	200	∞
	150	4000	5200	Depth in m.

M.T.P. No. 4...
FILE No. .....
JOB No. M.T./665/69.(7687)

LOCATION: SHARSHERET
COORDINATES: 109160 — 087430
ELEVATION: 91 m.
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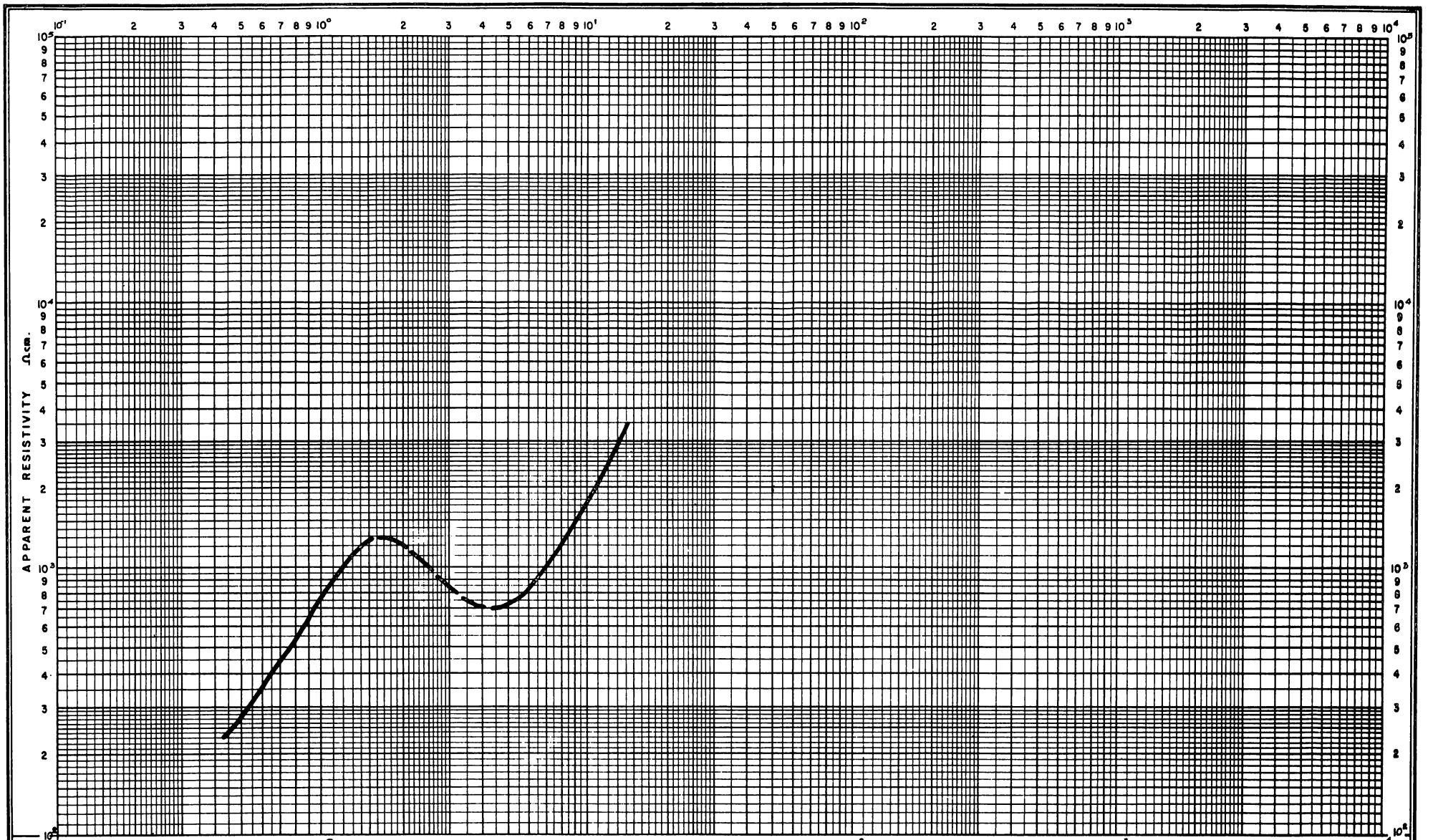
CALC. RES. Ωcm.	400	4000	400	100000
Depth in m.	280	2520	4300	

M.T.P. No. 5 . . .
FILE No. . . . . .
JOB No. M.T. 665/69 (7688)

LOCATION: SHARSHERET
COORDINATES: 108 415 — 088 735
ELEVATION: 103 m.

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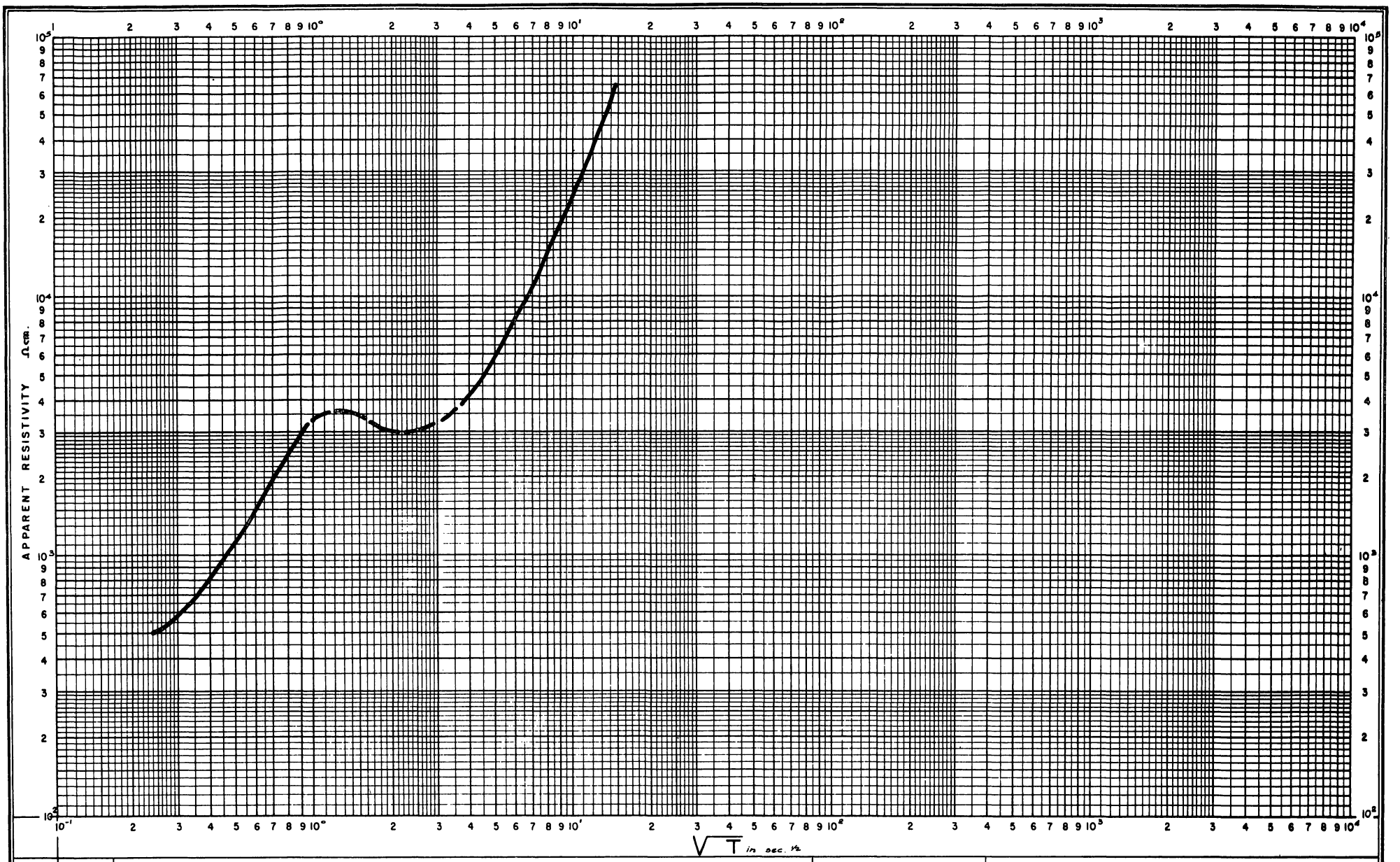


CALC. RES. Ωcm.	230	11500	230	∞
	250	2750	4400	Depth in m.

M.T.P. No. 6 . . .
FILE No. . . . . .
JOB No. MT/565/69(7689)

LOCATION: SHARSHERET
COORDINATES: 107100 — 089850
ELEVATION: 105 m.

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CALC. RES. Ωcm.	600	120000	600	∞
	310	4350	5350	depth in m.

M.T.P. No. 7. . .

FILE No. . . . . .

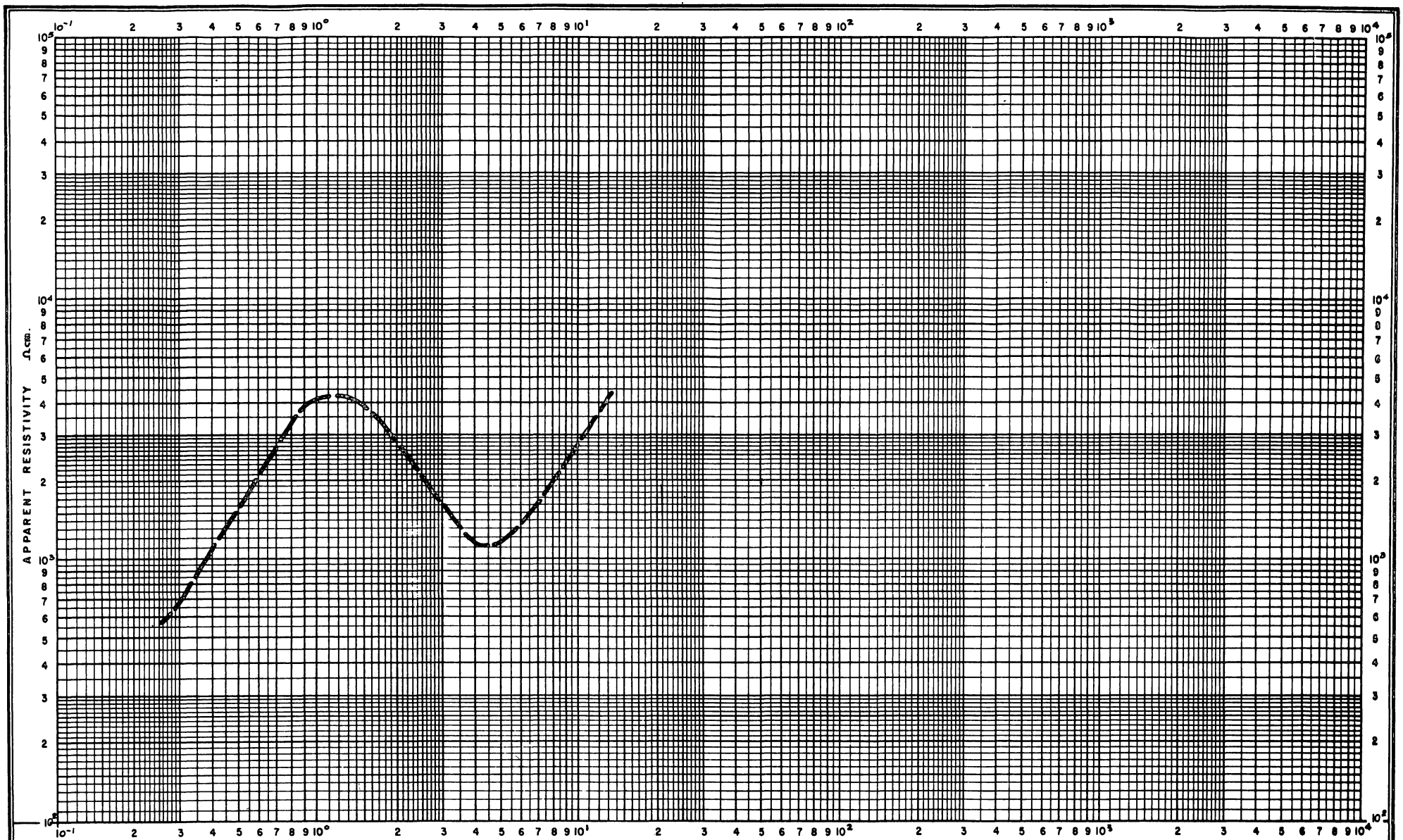
JOB No. MT, 665, '69 (7690)

LOCATION: GERAR

COORDINATES: 117200 — 089650

ELEVATION: 140 m.

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CALC. RES. Ωcm.	600	60000	200	∞
	850	4000	5200	Depth in m.

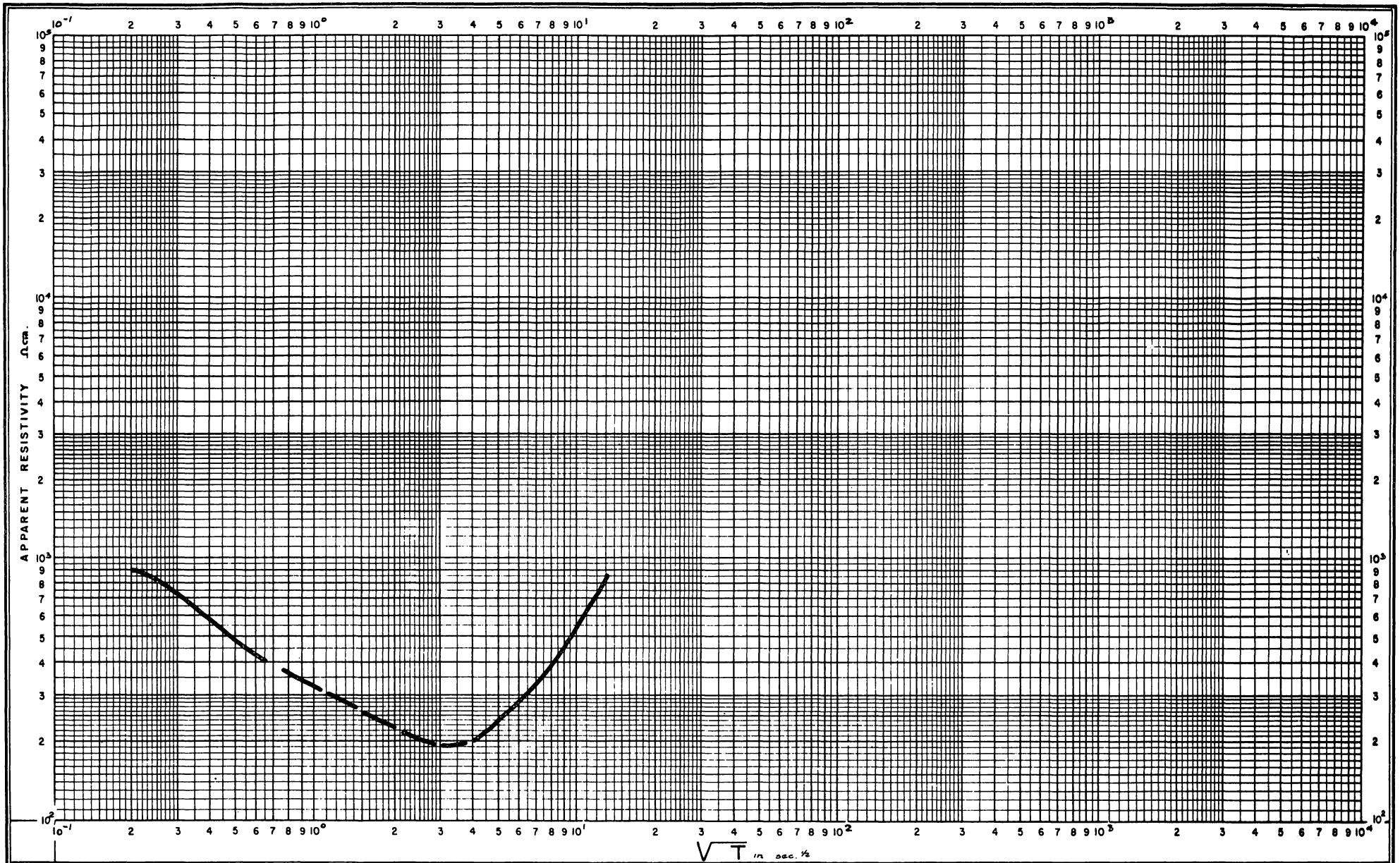
M.T.P. No. 8 . . .
FILE No. . . . . .
JOB No. MT/665/69(7691)

LOCATION: GERAR

COORDINATES: 116600 — 091600

ELEVATION: 155 m.

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CALC. RES. Ωm.	900	200	22500
	840	2650	Depth in m.

M.T.P. No. 9

FILE No. . . . . .

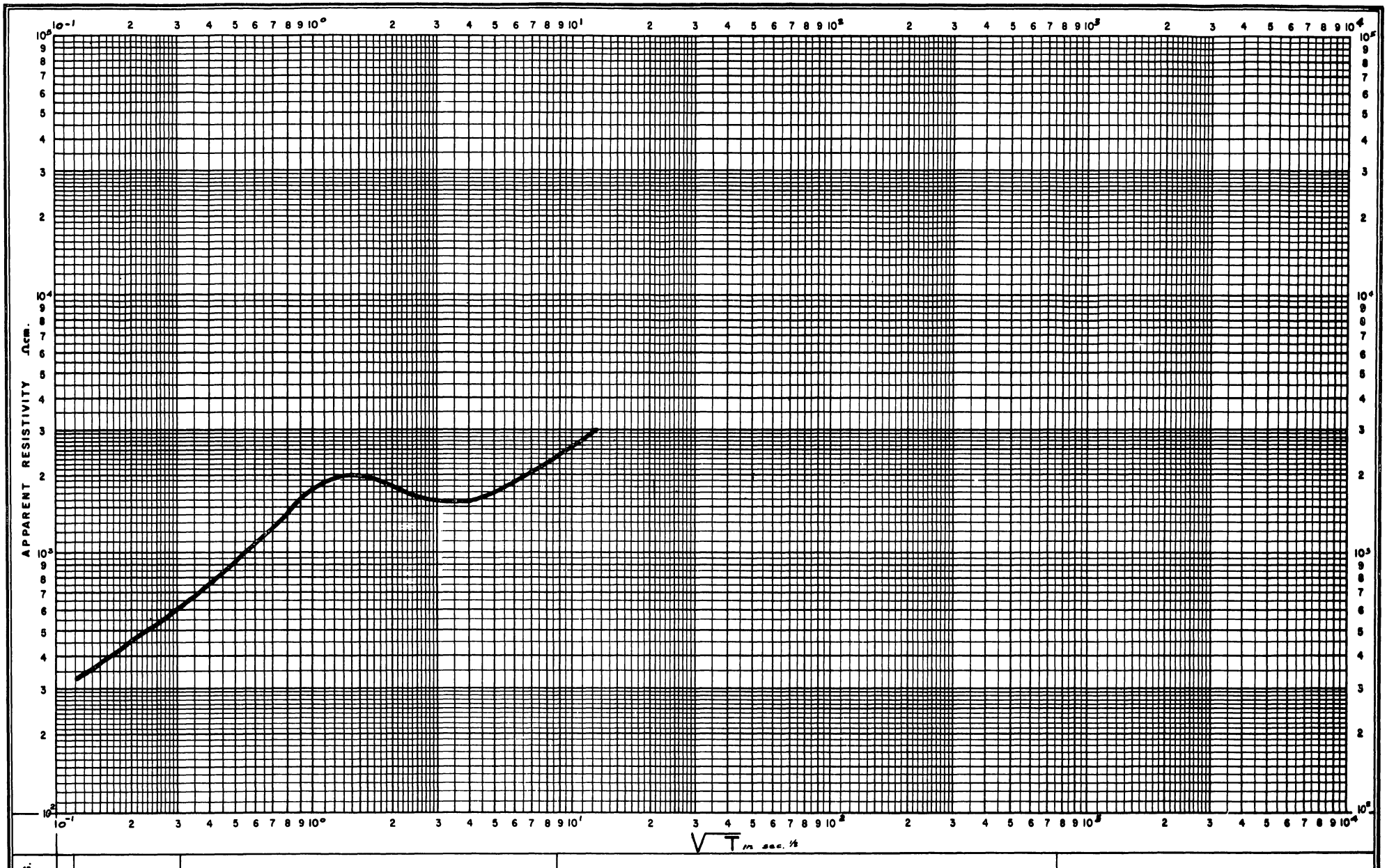
JOB No. M.T./065/69.(7692)

LOCATION: AFIQ

COORDINATES: 103386 — 094304

ELEVATION: 87 m.

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M.T.P. No. 10

FILE No. . . . . .

JOB No. MT/665/(7693)

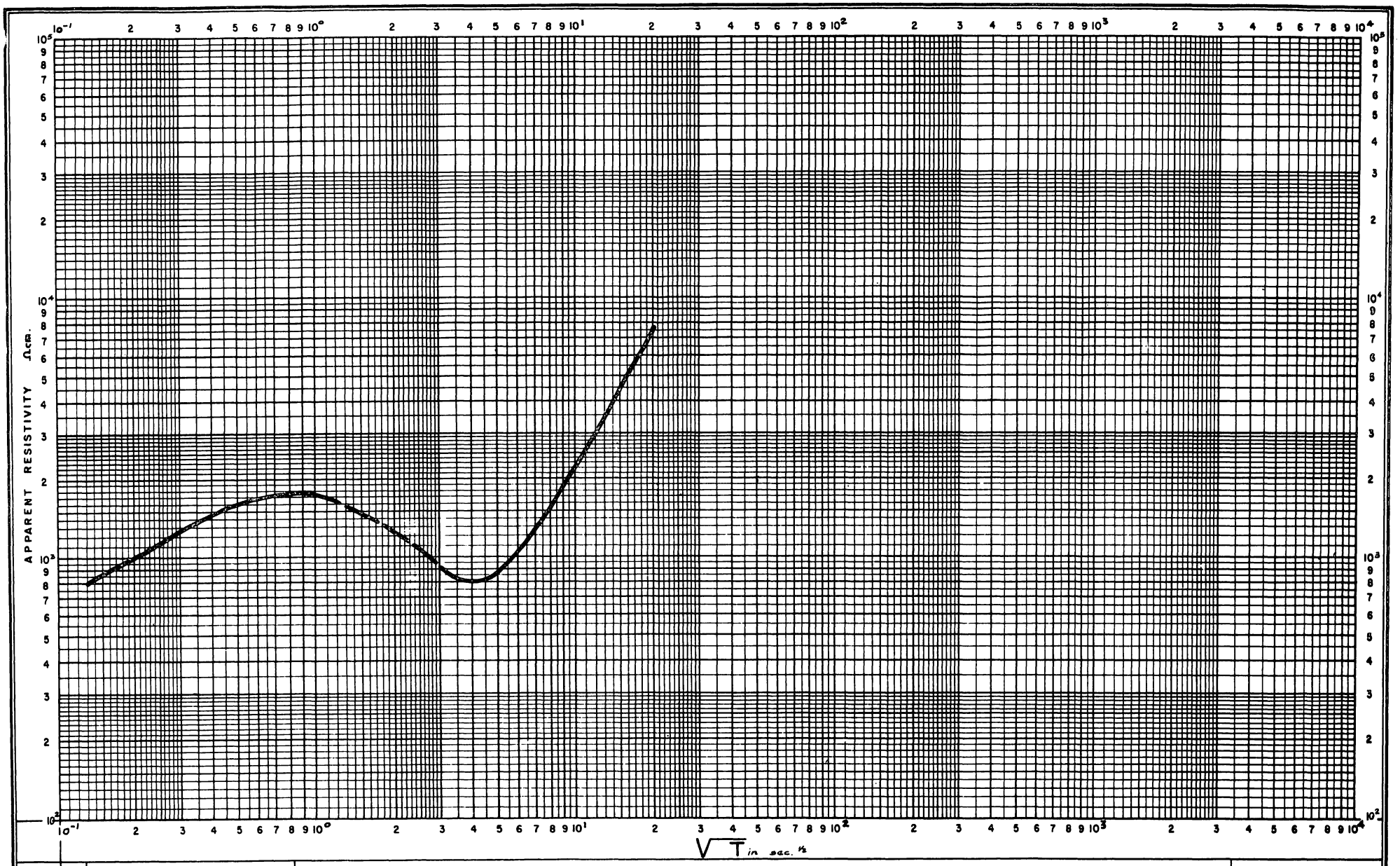
LOCATION: SHOVAL

COORDINATES: 125 750 — 088 700

ELEVATION: 225 m.

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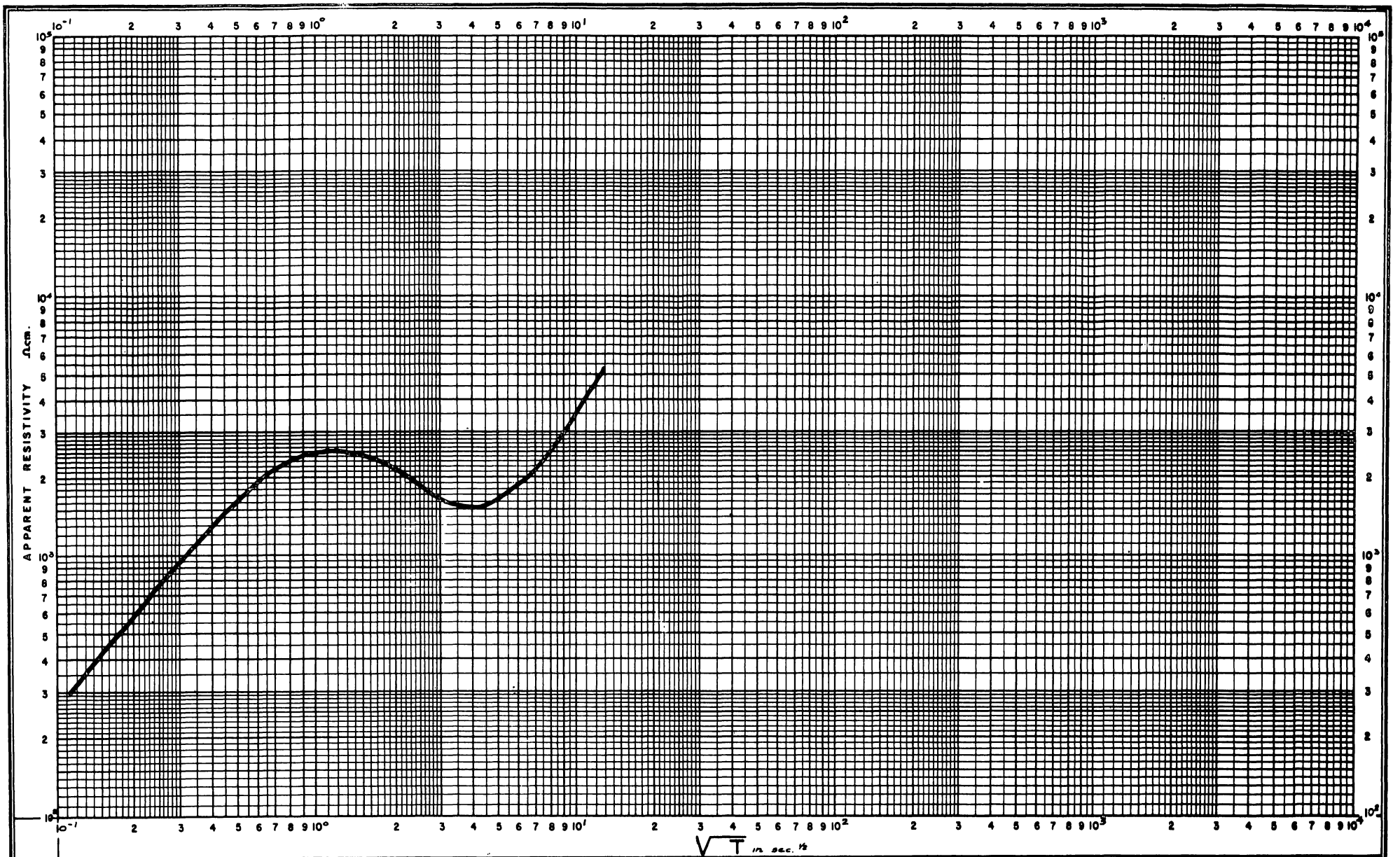
G-181



CALC. RES. Ωcm.	800	2400	800	∞
	150	1350	6750	Depth in m.

M.T.P. No. 11 . . .	LOCATION: ZIQLAG	
FILE No.	COORDINATES: 135 940 — 090 555	ELEVATION: 340 m.
JOB No. M.T./665/69 (7694)	THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS - ISRAEL -	

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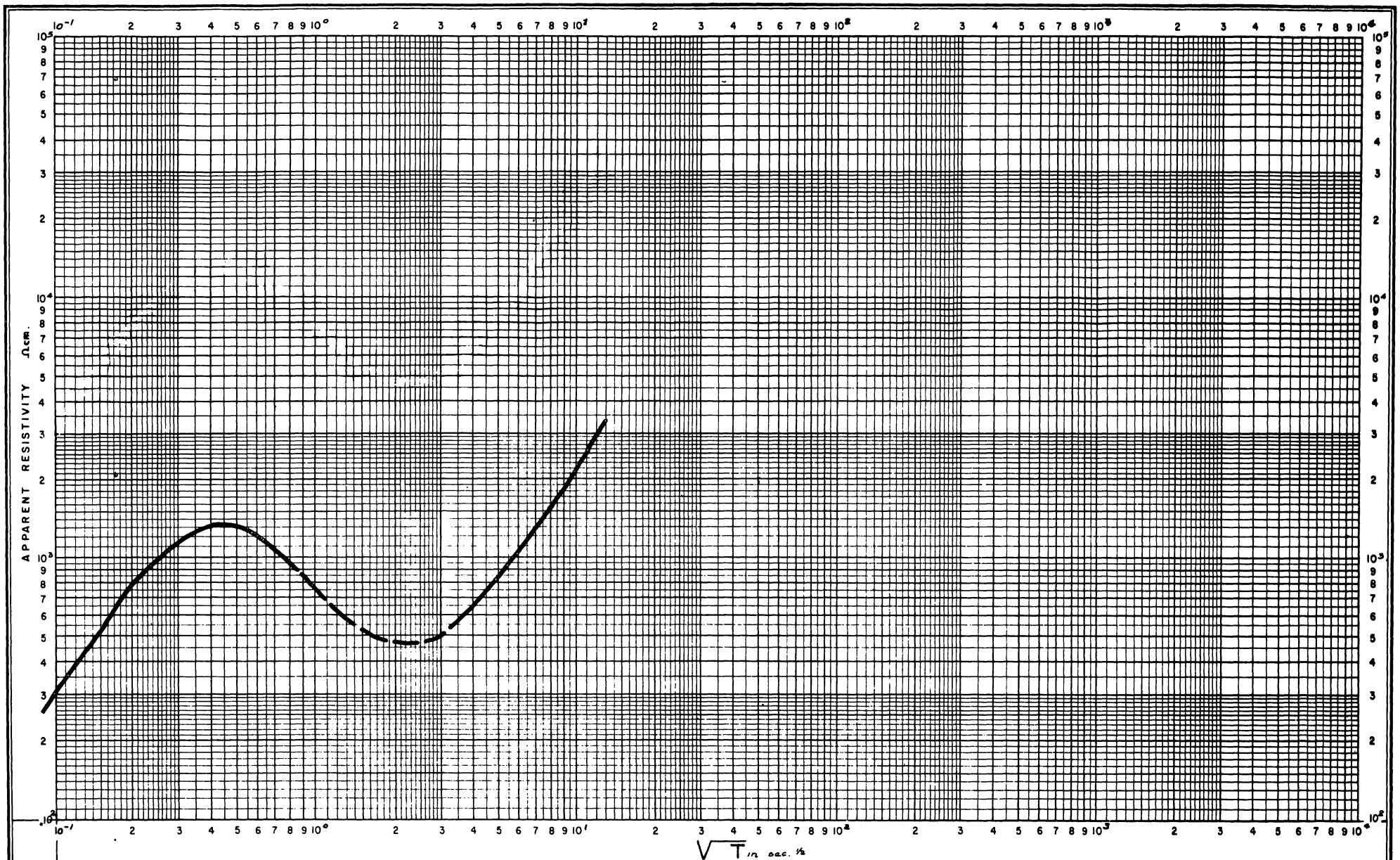


CALC. RES. Ωcm.	320	5500	1500	137500
	55	1700	8500	Depth in m.

M.T.P. No. 12 . .
FILE No. . . . . .
JOB No. MT/665/69.(7695)

LOCATION: LAHAV
COORDINATES: 139100 — 080750
ELEVATION: 370 m.

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CALC. RES. Ωcm.	240	12000	240	12000
depth in m.	60	660	2000	

M.T.P. No. 13

FILE No. .....

JOB No. M.T./665/69.(7697)

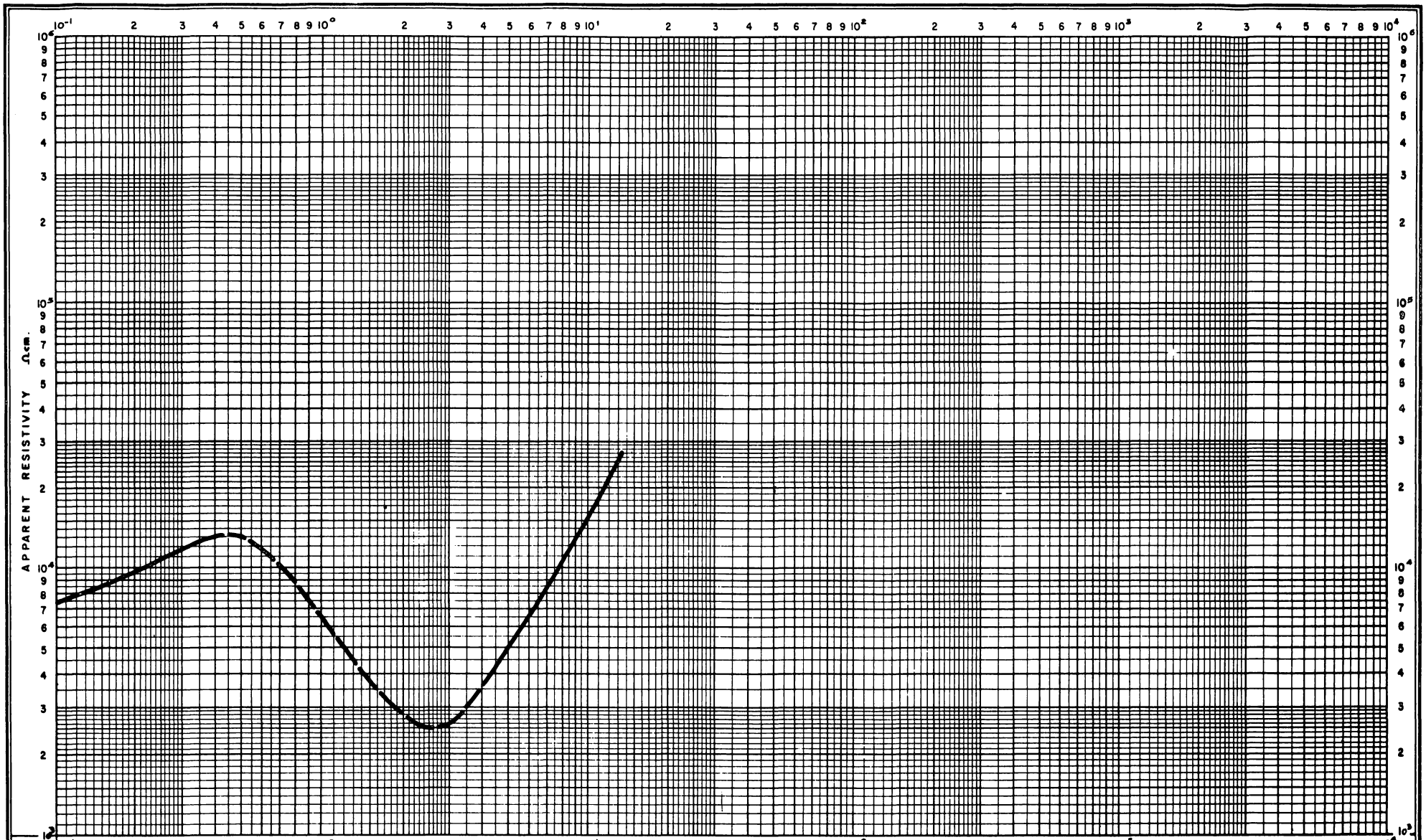
LOCATION: SHOKET

COORDINATES: 142 800 — 075 700

ELEVATION: 370 m.

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6.181



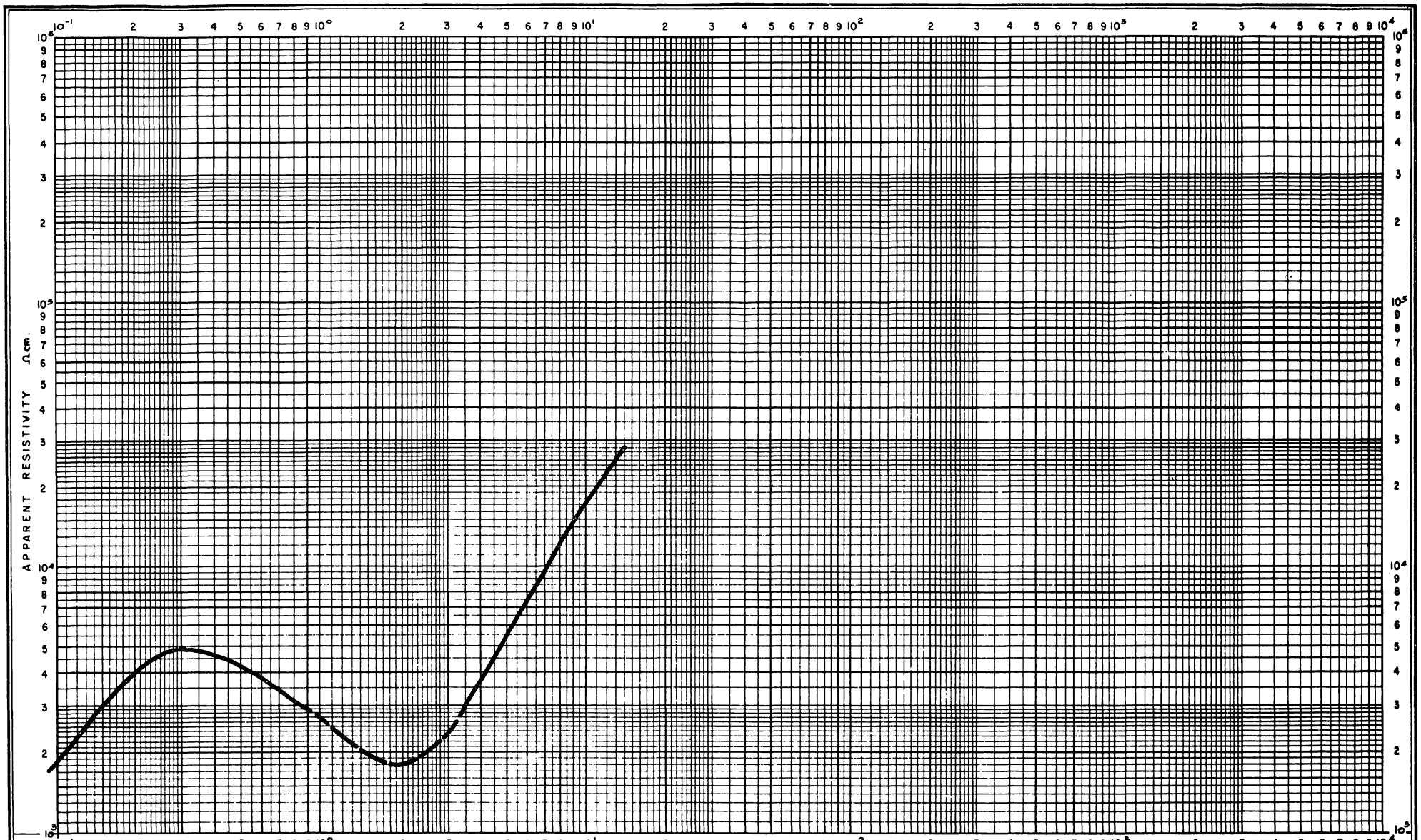
CALC. RES. Ωcm.	7500	15000	750	375000
	360	2600	4500	Depth in m.

M.T.P. No. 14..
FILE No. .....
JOB No. MT/.665/69 (7697)

LOCATION: SHOKET
COORDINATES: 149 800 — 074 200
ELEVATION: 485 m.

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CALC. RES. Ωcm.	1600	23300	1600	∞
	150	900	4500	Depth in m.

M.T.P. No. 15

FILE No. . . . . .

JOB No. M.T/665/69 (7698)

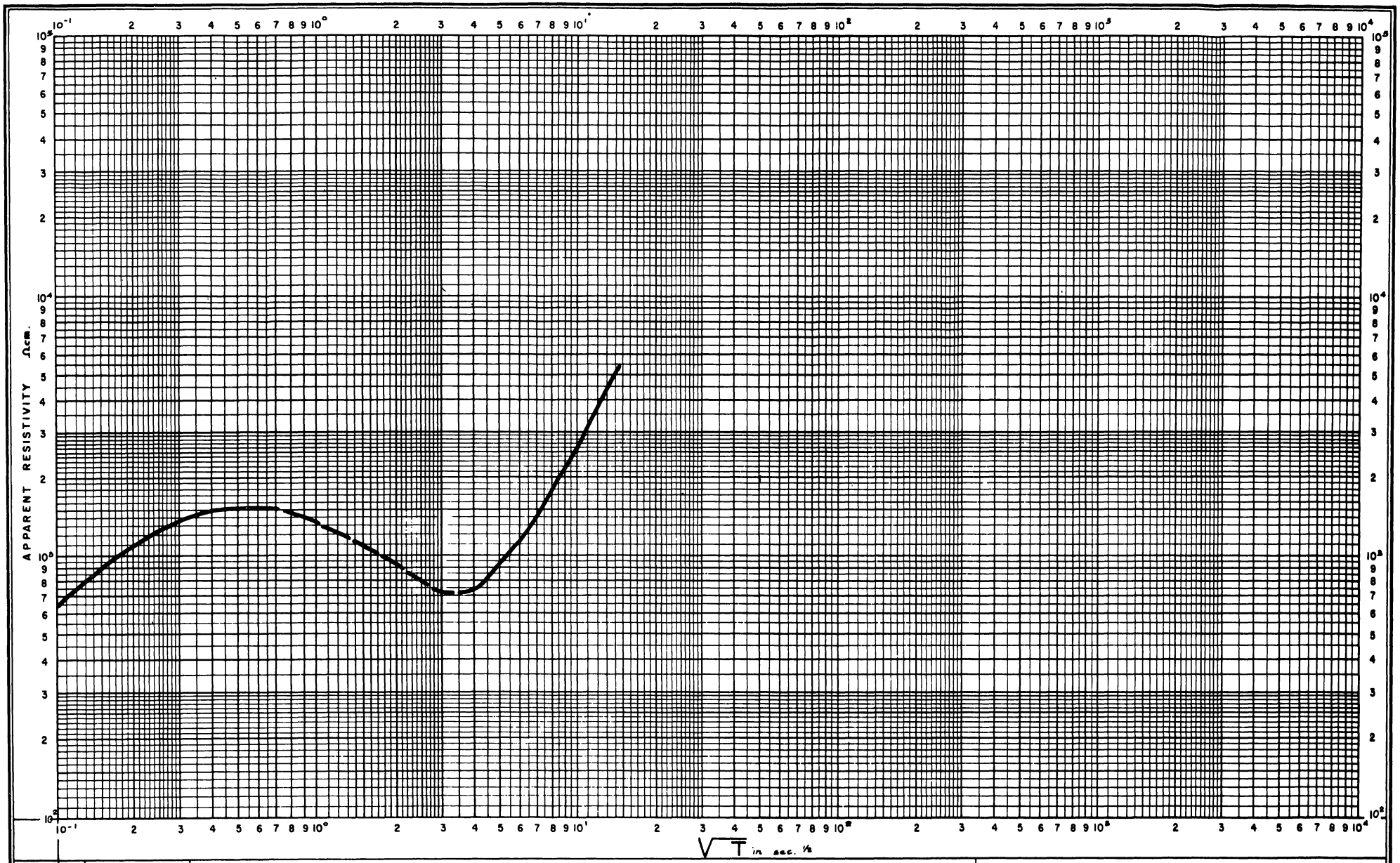
LOCATION: KASIF

COORDINATES: 156 300 — 072 750

ELEVATION: 475 m.

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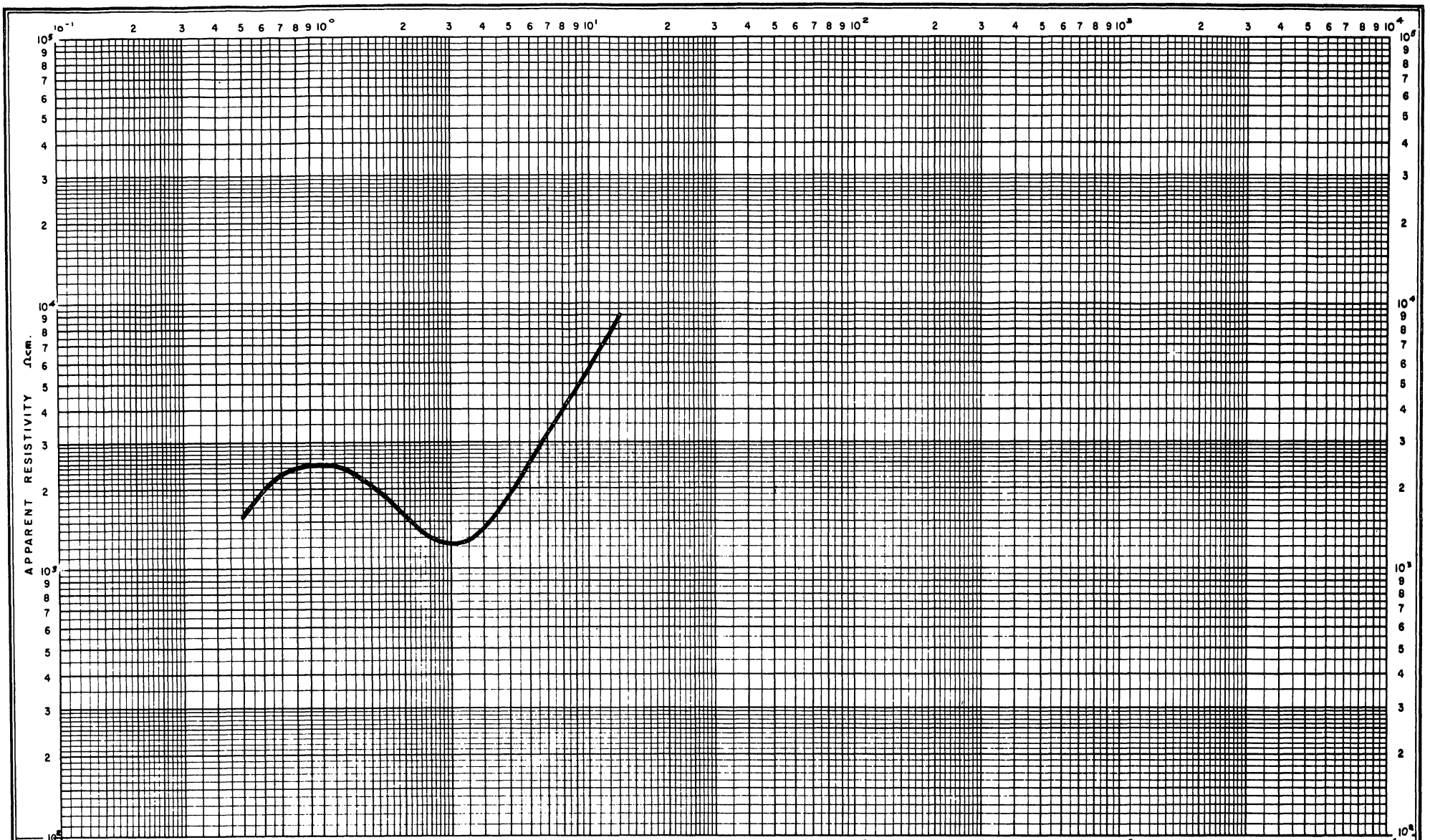
CALC. RES. Ωm.	700	7000	700	∞
	150	750	5250	Depth in m.

M.T.P. No. 16. . .
FILE No. . . . . .
JOB No. MT/665/69.(7699)

LOCATION: ARAD
COORDINATES: 164 850 — 074 900
ELEVATION: 510m.

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e.191



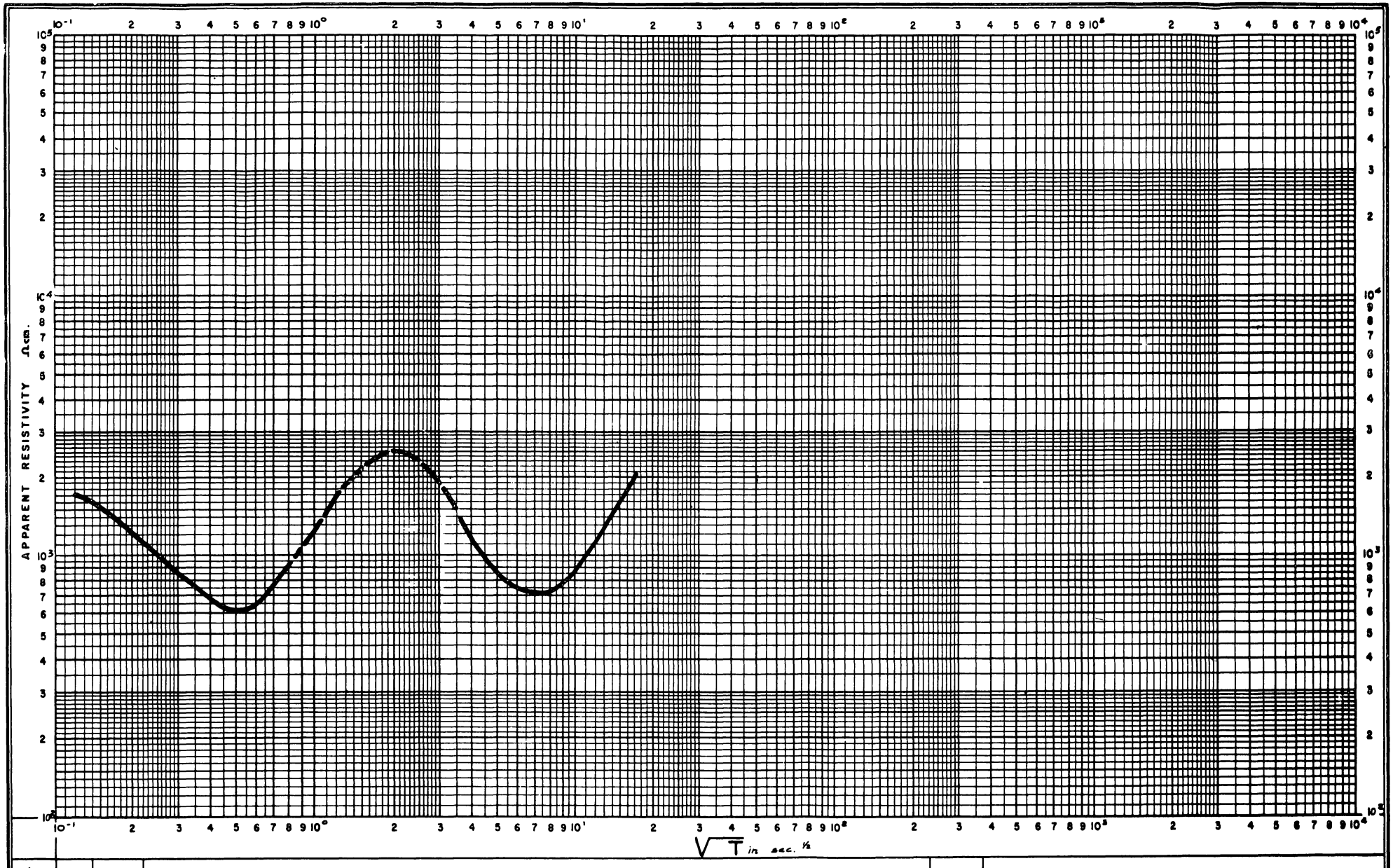
CALC. RES. Ωcm.	1500	22500	300	337500
	770	2310	3400	Depth in m.

M.T.P. No. 17. . .
FILE No. . . . . .
JOB No. MT/665/69 (7700)

LOCATION: TAMRUR
COORDINATES: 180 850 — 065 250
ELEVATION: 105 m.

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6.161

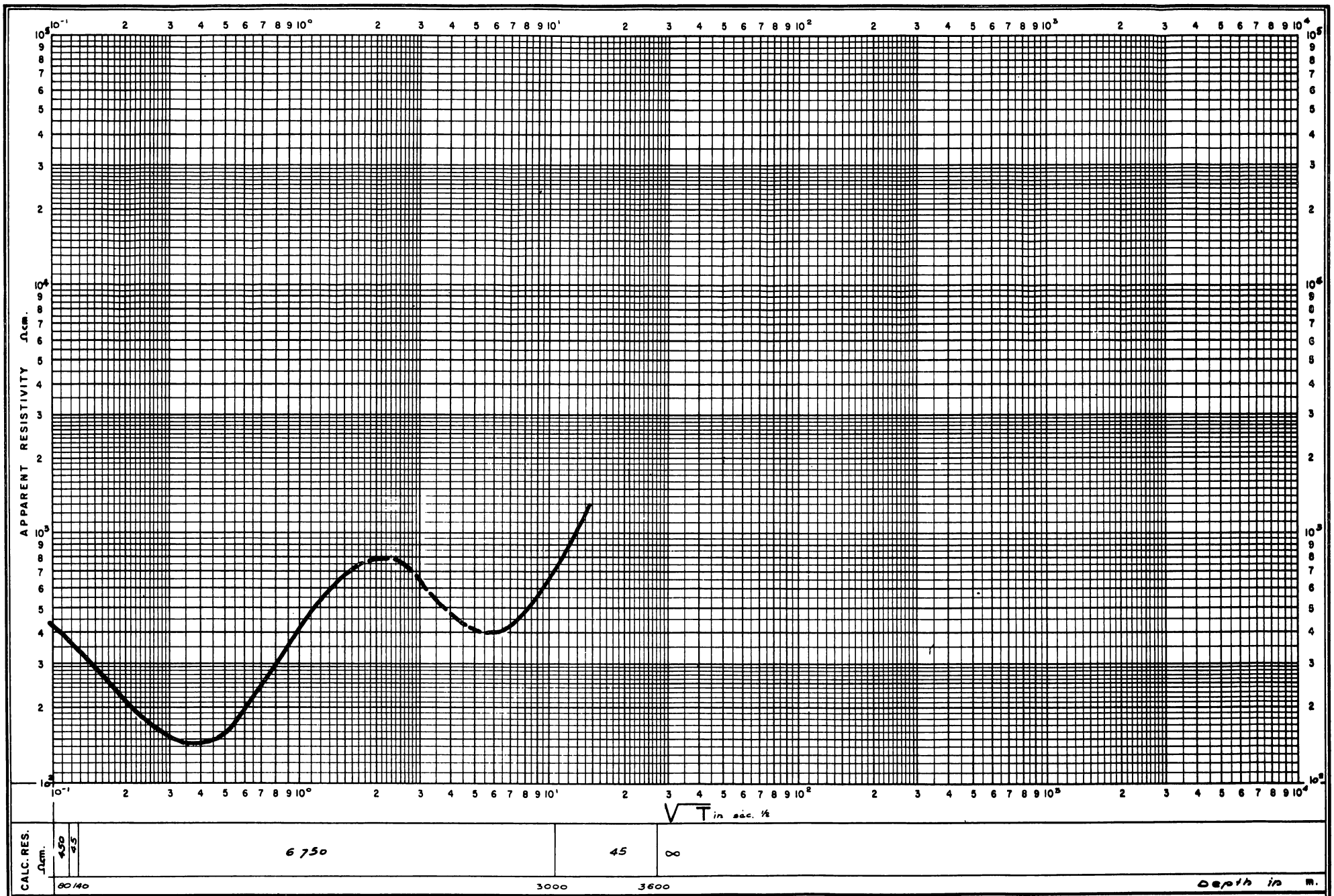


CALC. RES. Ωcm.	1700	340	42500	30	∞
depth in m.	200	500	5000	5300	

M.T.P. No. 19. . .
FILE No. . . . . .
JOB No. MT/.645/69.(7701)

LOCATION: KIDOD
COORDINATES: 171900 — 076750
ELEVATION: 600 m.

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M.T.P. No. 20..

FILE No. .....

JOB No. MT/665/69(7702)

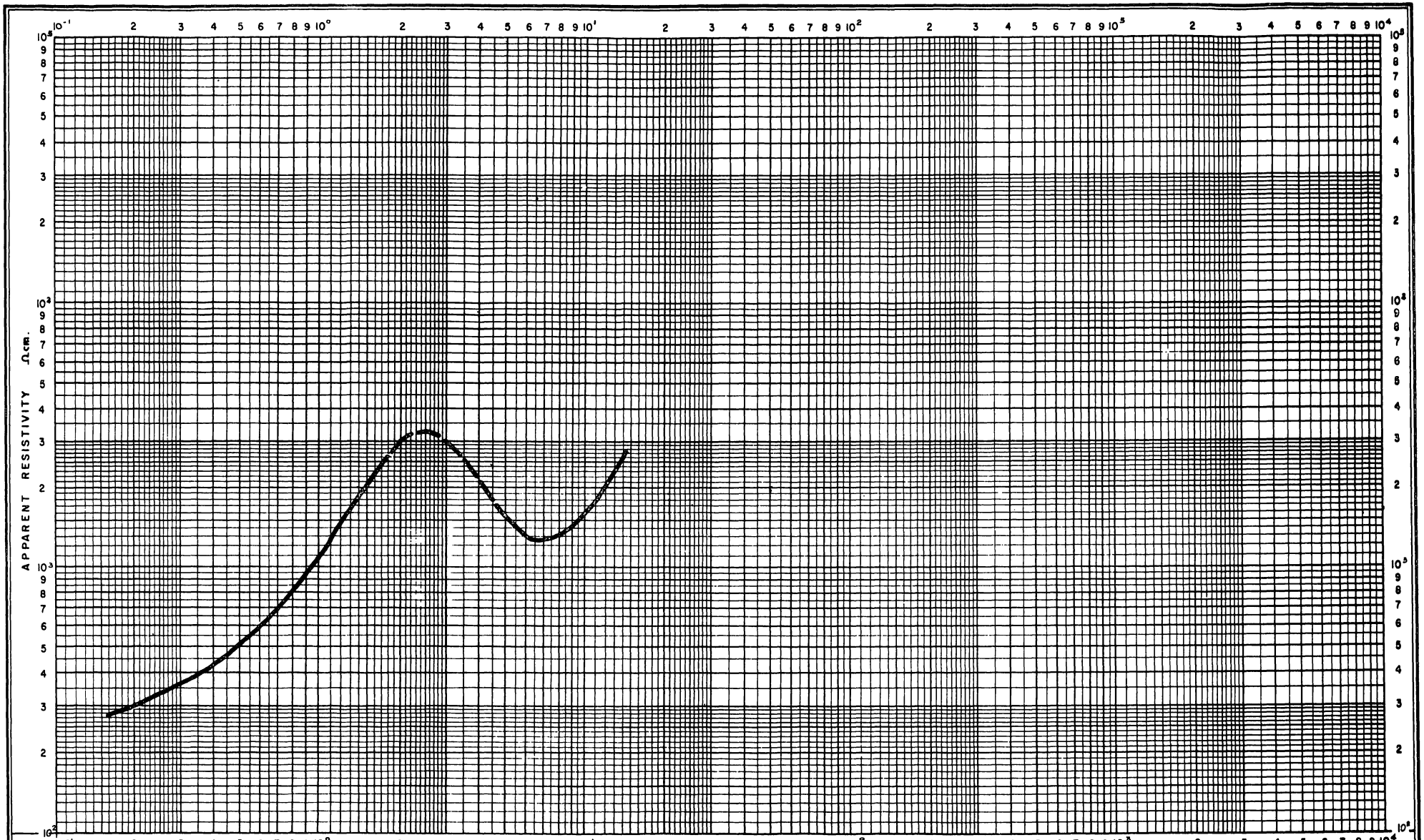
LOCATION: LOT

COORDINATES: 185300 — 059650

ELEVATION: — 360 m.

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0.188



CALC. RES. Ωm.	300	1500	30000	60	∞
	140	1000	7000	7500	Dept in m.

M.T.P. No. 21

FILE No. .....

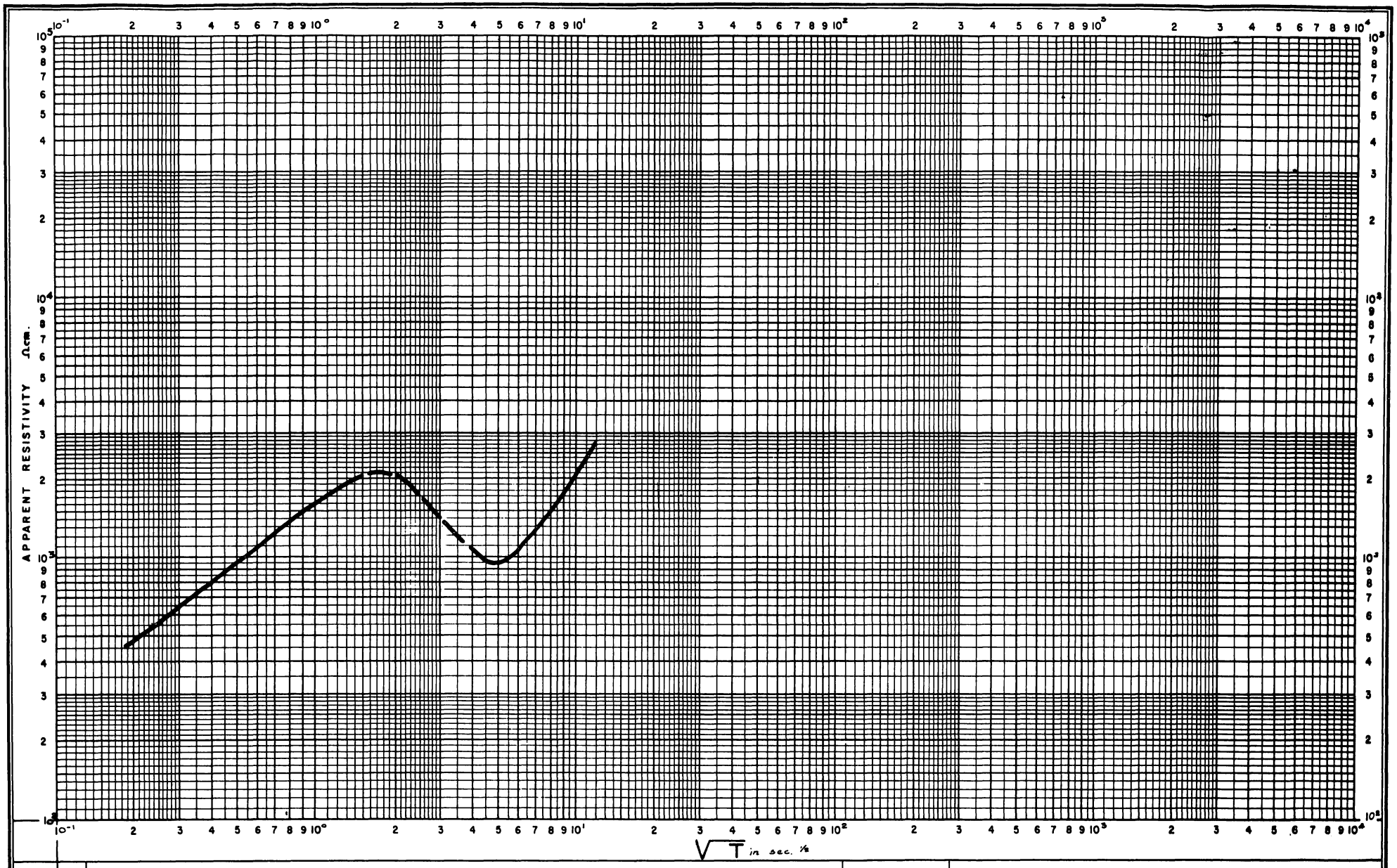
JOB No. MT/665/69 (1703)

LOCATION: SHOVAL

COORDINATES: 121150 — 090450

ELEVATION: 170 m.

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CALC. RES. Ω.cm.	450	2700	100	∞
	160	4500	5100	Depth in m.

M.T.P. No. 22

FILE No. .....

JOB No. MT/665/69.(7704)

LOCATION: LAHAV

COORDINATES: 132 050 — 090 000

ELEVATION: 290m.

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e.181